



Context Labs

TRACE-H METHODOLOGY VERSION 2.5

**Traceable Reconciled Attribute & Carbon Emissions
for Hydrocarbons Methodology**

May 2026

[Abstract](#)

This document details Context Labs methodology for lifecycle emissions accounting across the natural gas value chain, for the generation of Asset-Grade Environmental Attributes.

Table of Contents

1 INTRODUCTION	1
2 REFERENCED PROTOCOLS.....	3
3 SUMMARY DESCRIPTION OF THE TRACE-H METHODOLOGY	4
4 DEFINITIONS	6
5 APPLICABILITY CONDITIONS	8
6 PROJECT BOUNDARY	8
7 DATA INPUTS AND USES	8
7.1 FACILITY OPERATIONAL DATA	8
7.2 EMISSIONS MONITORING DATA	9
8 DATA MANAGEMENT	10
9 DATA QUALITY LEVELS & EXCEPTION REPORTING.....	10
9.1 DATA QUALITY LEVELS	10
9.2 EXCEPTION REPORTING	11
10 BOTTOM-UP SOURCE-LEVEL EMISSIONS CALCULATIONS	11
10.1 PRIMARY EMITTING FACILITIES – SCOPE 1	12
10.2 PRIMARY EMITTING FACILITIES – SCOPE 2	19
10.3 MINOR EMITTING FACILITIES	19
10.4 CO ₂ EQUIVALENCE	22
10.5 STATISTICALLY REPRESENTATIVE SAMPLING.....	22
11 TOP-DOWN, SITE-LEVEL MEASUREMENTS AND SITE EMISSIONS RECONCILIATION.....	24
12 EMISSIONS ALLOCATION METHOD FOR END-TO-END EMISSIONS TRACKING	28
12.1 PRODUCTION SEGMENT	28
12.2 MIDSTREAM SEGMENT.....	29
12.2.1 Construct a Network Graph of End-to-End System	30
12.2.2 Identify the Emitting Facilities Along Each Scheduled Flow Path	31
12.2.3 Determine the Total Throughput and Emissions per Unit for Each Emitting Facility.....	31
12.2.4 Determine Emissions per Unit for Pipeline Segments.....	35
12.3 NATURAL GAS FIRED POWER GENERATION FACILITIES	35
12.4 AGGREGATE EMISSIONS ACROSS THE FLOW PATH.....	36
12.5 NATURAL GAS LIQUIDS.....	37
13 VERIFICATION	38
14 REFERENCES	49
APPENDIX A – ABOUT THE CONTRIBUTING ORGANIZATION	51
APPENDIX B – REVISION LOG.....	54

List of Figures and Tables

Figures	
Figure 1	Illustrative overview of the Context Labs emissions quantification methodology
Figure 2	Contextual Data Fabric – Core Components and Features
Figure 3	Reconciliation Process Flowchart
Figure 4	Map view of Transco pipeline network graph with >1800 locations connected by pipeline segment
Figure 5	Commercial nominations vs. actual physical gas flows
Figure 6	Emissions aggregation across the end-to-end flow path
Tables	
Table 1	Bottom-up methane emissions monitoring and quantification summary
Table 2	GWP100 Factors
Table 3	GWP20 Factors
Table 4	Statistically Representative Sample Size Requirements (modelled from OGMP 2.0)
Table 5	Sample locations for Transco Mainline sorted for Milepost
Table 6	NGLs example energy allocation
Table 7	Example Pathway Intensities and Total Emissions

1 Introduction

A key challenge to managing greenhouse gas (GHG) emissions is accurately understanding the specific quantity and distribution of those component emissions (primarily CO₂, CH₄, and N₂O for natural gas related systems) into the atmosphere. Methane emissions across the natural gas value chain from well head to gas-fired power production occur during normal operations (e.g., continuous bleeds from pneumatic devices and methane slip) or planned maintenance (e.g., depressurizing a piece of equipment for maintenance), or can also result from abnormal events (e.g., leaks, equipment malfunctions, system upsets, or damage caused by third parties). These emissions (notably those from unplanned events) can be particularly challenging to quantify as the onset of an emissions event and the methane release rate are often undetectable using traditional process and safety monitoring instrumentation found in common operating facilities. Uncertainty regarding the duration, frequency, and release rates results in uncertainty in the total quantity of methane released. While carbon dioxide may be present in small quantities in certain gas release events and sources, CO₂, along with N₂O, is primarily generated through combustion-related activities. These emissions are accounted for within the relevant source categories identified throughout this document.

Although many methods for GHG emissions estimation exist, conventional approaches rely heavily on laboratory-developed heuristic models, sometimes referred to as ‘emissions factor models.’ While these emissions factor models have the advantage of being simple for all companies to apply consistently, they fail to account for site-specific factors that can have significant effects on actual emissions—such as the equipment manufacturer and model, commissioning practices, time in operation, or maintenance programs. Many of the emissions factors defined within the U.S. EPA Greenhouse Gas Reporting Program (GHGRP) originate from dated studies, some almost three decades ago through a study by the Gas Research Institute released in 1996. According to a more recent publication by the National Academies of Science, Engineering & Medicine, in many cases, these emissions factors do not represent the current population of emissions sources, facility designs, or work practices in place today.¹

Recent technological advances have greatly improved methane sensing instrumentation, but there remains no single sensor with both the high sensitivity to detect low-rate fugitive emissions and the broad swath to continuously monitor all possible leak points across a large, distributed system. It is critical to ingest and integrate disparate data from multiple measurement sources to deliver a more comprehensive and accurate view of methane emissions based on the best available measurement for the facility. The TRACE-H methodology improves on previous (especially emissions factor) approaches by integrating:

- Multiple conventional and advanced sensing and quantification devices for comprehensive and cost-effective emissions identification and measurement
- Direct measurement-informed quantification
- Rigorous data-driven analyses
- Robust and holistic emissions monitoring, mitigation, and reporting programs for operating companies and stakeholders

The basic shared approach of current greenhouse gas emissions estimation methods involves the identification of potential GHG-emitting sources at an operating facility. These sources include

continuous emission sources, as well as discontinuous or intermittent sources and events. With appropriate source-level measurements, routine emissions from these common sources can be reasonably determined using engineering calculations combined with periodic measurements and site operating data. Common GHG emissions sources located across the natural gas to energy value chain include:

- Combustion Engines and Turbines
- Reciprocating & Centrifugal Compressor Vents
- Reciprocating Engine Crank Case Venting
- Gas starters
- Atmospheric Storage Tanks
- Gas-Driven Pneumatic Devices
- Desiccant Dehydrators
- Glycol Dehydrators
- Acid-Gas Removal Units
- Well Venting for Liquids Unloading
- Casinghead or Annulus Vent
- Component Leaks
- Blowdowns
- Purges
- Control devices (e.g., flares, thermal oxidizers, and combustors)
- Electricity Usage (Scope 2)

Several studies have demonstrated that a significant fraction of methane emissions at a facility can originate from a small fraction of sources.¹ High-emitting sources generally fall into three categories:

- Chronic high-emitting sources:
 - Do not vary over time and persistently emit at higher rates compared to similar sources
 - Typically caused by poor design or operational practice
- Episodic high-emitting sources:
 - Often associated with known, intentional releases such as blowdowns, equipment evacuation for maintenance, or liquids unloading
 - Produce very high methane emissions rates, potentially over very short periods of time (e.g., minutes)
- Malfunctioning high-emitting sources:
 - May be intermittent or prolonged
 - Examples include malfunctioning pneumatic devices, stuck-open dump valves allowing gas blowby to a downstream low-pressure tank, and leaking compressor isolation and blowdown valves.

The use of advanced detection and measurement technologies within this methodology helps to more rapidly identify the presence of high-emitting sources and unexpected events in order to perform timely remediation. In addition, these technologies assist in more accurately quantifying reconciled whole-site emissions and improving confidence in reported emissions values.

2 Referenced Protocols

The approach to emissions quantification set forth in this methodology has been developed based on the guidelines outlined in industry-leading protocols, including the Oil and Gas Methane Partnership (OGMP) 2.0 Reporting Framework and the Gas Technology Institute (GTI) Veritas Protocol Version 2.0. In accordance with the GTI Veritas Protocol, this methodology considers the minimum detection threshold and uncertainty of sensing technology to leverage advanced measurement technology while ensuring comprehensive coverage and works to reconcile top-down and bottom-up emissions methods. In accordance with OGMP 2.0, in addition to the features listed above, this method implements emissions reported by detailed source type and specific emissions and activity factors based on source-level measurement (Level 4), with >90% of the expected methane emissions sources from each site meeting Level 4 requirements for source-level testing. Site-level measurements with risk-based facility revisit frequency (Level 5) are currently being performed at least semi-annually.

Gas Technology Institute (GTI) Veritas Protocol

- Emissions intensity protocol
- Measurement & reconciliation protocol for Upstream
- Measurement & reconciliation protocol for Midstream
- Audit and assurance protocol

Oil and Gas Methane Partnerships (OGMP) 2.0

- Benchmarking
- Emission reduction approaches
- Technology advancement
- Policy development
- Specific, measurement-based emissions and activity factors (Level 4) for source-level emissions estimation
- Top-down, site-level measurement (Level 5)

API Compendium of Methane Emission Estimation Techniques

US EPA's Greenhouse Gas Reporting Program (GHGRP)

- Established, objectively defined engineering calculations for bottom-up emissions estimation
- Combined with higher-frequency, unit/site-specific measurement-based emissions and activity factors

Natural Gas Sustainability Initiative's Methane Emissions Intensity Protocol

- Emissions intensity calculations
- Allocation of methane emissions to gas and liquid hydrocarbon product streams

3 Summary Description of the TRACE-H Methodology

The TRACE-H methodology was initially developed by Context Labs, B.V. in 2023 to enhance the accuracy of quantification of methane and total greenhouse gas emissions. It was built upon the protocols referenced in Section 2 above and it provided a scientific basis for emissions quantification that was deployed through Context Labs' platform. In 2024, Context Labs assigned the copyright of the methodology (version 2.4) to the Open Decarbonization Project (ODP) under the Massachusetts Institute of Technology (MIT) open-source license. This updated version 2.5 has incorporated inputs from Williams, EQT, and Jonah Energy as an open-source methodology.

To support a holistic and comprehensive approach to emissions quantification, the methodology incorporates external emissions measurement and monitoring from a variety of conventional (OGI cameras, Hi-Flow Samplers, exhaust stack testing) and advanced (Hyper/multi-spectral cloud imaging or LDAR-based near-Continuous Emissions Monitoring Systems, LiDAR-equipped aerial flyover surveys, and Advanced methane detection satellite technologies) methane sensing devices for direct measurement-informed quantification of emissions across the full natural gas to power value chain. Measured emissions data are combined with real-time operational data from process control systems and operators' logbooks to quantify emissions on an ongoing basis.

Data veracity and confidentiality are foundational to the data management approach underpinning this methodology. All data used in emissions quantification should maintain an immutable, auditable, end-to-end record of provenance to irrefutably prove the data has never been falsely represented, manipulated, or otherwise corrupted. At each point where data is created, transformed, combined, analyzed, or accessed, a 'Digital Proof' should be created and published immutably to a distributed ledger (i.e., blockchain) to track and immutably preserve data pedigree and provenance for audit or verification purposes. Recognized third-party verifiers and subject matter experts act as 'Digital Notaries' submitting digitally signed statements attesting to the origin, integrity, and accuracy of raw data inputs or analytics outputs. This approach ensures the highest possible degree of trust, security, useability, and accountability related to emissions performance data.

The methodology is described in detail in Sections 7–12.

Section 7: Data Inputs and Uses

- Facility Operational Data
- Emissions Monitoring Data

Section 8: Data Management

- Ensure data is traceable and supported by documented lineage across the data lifecycle
- Use validated data with controls that maintain integrity and enable auditability and independent verification

Section 9: Data Quality Levels & Exception Reporting

- Increasing reliance on measurement to reduce uncertainty
- Transparent disclosure of data quality level and any identified exceptions
- Incentivize continuous improvement, noting practical challenges of real-world operations

Section 10: Bottom-Up Source-Level Emissions Calculations

- Pre-deployment review of site equipment configuration and emissions inventories
- Higher frequency measurement of unit or site-specific emissions rates going beyond compliance requirements
- Measurement-informed engineering calculations using specific emissions factors and activity factors aligned with OGMP 2.0 Level 4

Section 11: Top-Down, Site-Level Measurements and Site Emissions Reconciliation

- Semi-annual aerial flyover surveys for top-down, site-level measurement
- Reconciliation of bottom-up emissions and site-level measurements aligned with OGMP 2.0 Level 5

Section 12: Supply Chain Emissions Allocation

- Allocate GHG emissions with gas and liquid hydrocarbon product streams
- Associate emissions with specific product shipments
- Adjust for system-wide emissions
- Trace emissions end-to-end across the supply chain

Section 13: Verification

- Rigorous standards for emissions attestation, based on 3rd-party review

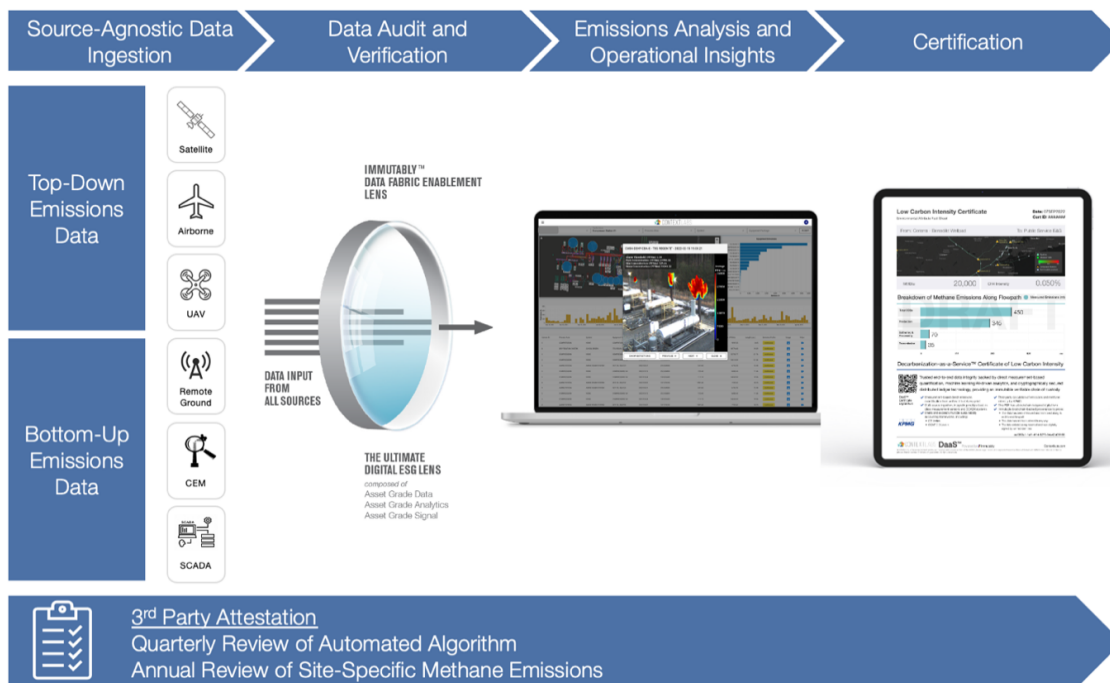


Figure 1. Illustrative overview of the Context Labs emissions quantification methodology

4 Definitions

Asset Hierarchy

A hierarchically organized list of all identifiable components (e.g., a valve, instrument, transmitter, etc.), equipment (e.g., a compressor), equipment groups (e.g., dehydration unit), facilities (e.g., a plant site), and groups of facilities (e.g., an operating division or region).

Asset Grade Data (AGD)

Data which has full provenance and veracity established.

Asset / Operating Unit

A logical business or operating unit as defined by the company (e.g. producing basins, regional assets, pipeline networks, gathering facilities, etc.). An asset may be comprised of one or more sites / facilities.

Emissions Source

A piece of equipment, device, or component within a process system that releases greenhouse emissions to the atmosphere.

Data Fabric

A data fabric platform provides a centrally managed means for integrating disparate data sources intelligently and securely, providing a consistent means to secure and access, as well as a unified semantic data layer to support business use cases.

Engine Stack Test

These are periodic tests performed on combustion engines per EPA regulation to ensure that the engines meet emissions performance standards for common pollutants (CO, CO₂, NO_x, CH₄).

Emissions Factors

A representative value that attempts to relate the quantity of a compound released to the atmosphere with an activity associated with the release of that compound. Emissions factors are developed from source test data, material balances and engineering estimates of acceptable quality and are assumed to be representative of long-term averages for all facilities in the source category.

Emissions Inputs

Data inputs required to calculate emissions. Examples of emissions inputs are volume of natural gas throughput, equipment run-time, and mole fraction of GHG gas. Emissions inputs are used in emissions calculations, such as those found in EPA's 40 CFR 98 Mandatory GHG Reporting Rule.

Flyover

Aerial gas-imaging LiDAR methane plume detection and emissions rate estimation.

Geographical Information System (GIS)

A system that connects facility and equipment data to a map.

Leak Detection and Repair (LDAR)

LDAR is a work practice used to detect equipment leaks from point-source fugitive emission components, such as valves, pumps, pressure relief valves (PRVs), and connectors. LDAR Programs are predominately managed in accordance with federal, state, or voluntary standards that define survey frequency, leak thresholds, and repair requirements. Examples of data include the dates that leaks were identified, types of equipment leaking, and repair confirmation dates.

Light Detection and Ranging (LiDAR)

LiDAR scans a scene with lasers and measures the reflection or absorption patterns of light to determine the distance of an object or concentration of a target gas. The term LiDAR in this method refers to plane-mounted Gas Mapping LiDAR that detects and maps out methane leaks from the air.

Machine Learning

Application of mathematical and programming techniques to data, refining processes such that the system improves over time based on the data available.

Near-Continuous Emissions Monitoring System (Near-CEMS)

EPA defines continuous emission monitoring systems (CEMS) as, “the total equipment necessary for the determination of a gas or particulate matter concentration or emission rate using pollutant analyzer measurements and a conversion equation, graph, or computer program to produce results in units of the applicable emission limitation or standard.” Within this method, the use of the term Near-Continuous Emissions Monitoring System (Near-CEMS) refers specifically to methane emissions sensors installed on a permanent (or semi-permanent) basis for low-latency detection of elevated emissions. Certain tower-mounted laser or hyper/multi-spectral optical Gas Cloud Imaging (GCI) cameras may offer quantification capabilities and can potentially be used to locate methane emissions to a specific equipment source. Other advanced near-continuous monitoring technologies may also be incorporated in the future.

Site / Facility

A collection of sources with some relation to one another within a process system. Typically, found within a contiguous piece of property at a discrete geographical location or physical address.

Supervisory Control and Data Acquisition (SCADA)

SCADA data allows companies to monitor and control equipment to improve operating efficiency. Examples of SCADA data include pressure, temperature, flow volume, and emissions (if available).

5 Applicability Conditions

This emissions quantification methodology applies to vented and fugitive greenhouse emissions for natural gas production, gathering, transportation, and processing facilities as well as NGL Fractionation and Natural Gas Fired Power Generation Facilities. Emissions are calculated on an equipment basis and are then aggregated to provide a site-, region-, and company-wide view for a particular operator. An emissions allocation methodology is provided for tracking and quantifying end-to-end supply chain emissions from the wellhead to the final delivery location.

6 Project Boundary

This methodology calculates greenhouse gas emissions for natural gas production, gathering and boosting, transportation, and processing facilities as well as NGL Fractionation and Natural Gas Fired Power Generation Facilities. This methodology addresses Total Scope 1 GHG emissions, including carbon dioxide, methane, and nitrous oxide emissions, and Scope 2 (indirect) GHG emissions. Emissions from construction and development activities are not included at this time, but may be added as a future update. Emissions are quantified across sources at a facility—from routine day-to-day operations to unplanned and unintentional releases. Emissions from pipelines and pipeline blowdowns are also addressed. For end-to-end emissions tracking across the supply chain, a flow path must consist only of assets that have been evaluated following this method.

7 Data Inputs and Uses

To support a holistic and comprehensive approach to emissions quantification, this methodology incorporates real-time facility operational data and emissions monitoring from a variety of conventional (OGI cameras, Hi-Flow samplers, and exhaust stack testing) and advanced (e.g., Gas Cloud Imaging CEMS cameras, tower-mounted lasers, or LiDAR equipped airplanes, Advanced methane detection satellite technologies, etc.) methane sensing devices. Facility data provides context to emissions measurements, which helps inform operators of equipment or facility conditions that may lead to different emissions patterns. Emissions measurements are then combined with real-time operational data from process control systems & operators' logbooks to quantify emissions on an ongoing basis.

7.1 Facility Operational Data

Detailed, real-time facility data are critical for understanding how fluctuations in day-to-day operation impacts emissions. Operational data includes site-wide performance data as well as equipment-specific data and can be sourced from multiple operations data systems:

- SCADA readings
 - e.g., Operating temperatures
 - Operating pressures
 - Process flow rates, fuel gas flow rates
 - Run times, operating modes etc.

- Equipment design specification sheets
 - e.g., Equipment manufacturer
 - Model number
 - Rated horsepower
 - Design throughput, etc.
- Operator logs
 - e.g., Compressor blowdown log, etc.

Operational data is also important for determining different types of operating conditions, such as:

- Normal operations
- Planned maintenance or routine shutdowns (e.g., equipment depressurization, venting, purging, etc.)
- Unplanned events (e.g., process upsets, equipment malfunctions, leaks, etc.)

7.2 Emissions Monitoring Data

In alignment with OGMP 2.0, this methodology quantifies emissions using bottom-up methods, which are compared and reconciled against top-down, site-level emissions. Site-level measurements attempt to identify unexpected or high-emitting sources not captured using bottom-up methods to assist in more accurately quantifying emissions and improving confidence in reported values. Based on the facility type, configuration, and expected emissions distribution, measurement method(s) are selected to provide comprehensive and cost-effective coverage of site emissions. Detection limits and confidence intervals for each data source are discussed with the technology vendors and quantified where possible.

- Bottom-up data
 - LDAR
 - Survey equipment components (valves, PRVs, open-ended lines, flanges, and threaded connectors), gas-driven pneumatic devices, and atmospheric storage tank vents using OGI camera or EPA Method 21 Survey
 - Identify methane leaks/sources of leaks visually using OGI camera or based on methane concentration readings using Method 21 survey
 - Direct measurement of methane emissions mass rate for certain sources
 - Hi-Flow Sampler testing performed concurrently with LDAR surveys, direct measurement of methane emissions mass rate for the following sources:
 - Centrifugal compressor wet seal and dry seal gas vents
 - Reciprocating compressor rod packing seal vent
 - Blowdown vent through-valve leakage
 - Fugitive emissions detected during LDAR survey from equipment components, malfunctioning pneumatic devices, etc.
 - Storage tank vents
 - Stack Testing
 - Performance testing on combustion engines to monitor pollutants including NOX, CO and VOCs.

- Performance testing of destruction and removal efficiency (DRE) for emissions control devices including enclosed combustion devices, flares and thermal oxidizers. .
- Top-down data
 - Aerial / flyover data
 - Semi-annual aerial or satellite surveys to capture top-down, site-level measurements
 - Screen for changes to baseline conditions throughout the year
 - Near-CEMS (rapid detection, with potential use as top-down measurement tool)
 - Near-Continuous Emissions Monitoring Systems (near-CEMS) for low-latency emissions detection. Field trials are in progress to evaluate the effectiveness of using certain near-CEMS technologies with quantification capabilities as a high-frequency top-down measurement source.

8 Data Management

To comply with this methodology, all data used in emissions quantification, allocation, and reporting shall be subject to rigorous data management practices that ensure transparency, traceability, and auditability across the full data lifecycle. Each data element must maintain a documented and verifiable lineage, including its origin, any transformations or calculations applied, and its use in downstream analyses or reporting outputs. Appropriate metadata shall be retained to describe the context, timing, source, and handling of the data.

Data inputs should be derived from reliable and representative sources and, where practicable, reconciled across multiple independent measurements or datasets to improve accuracy and reduce uncertainty. Validation and quality assurance processes shall be implemented, including review or attestation by qualified personnel and, where applicable, independent third parties.

Systems and controls must be in place to ensure data integrity, including mechanisms to detect unauthorized modification, preserve version history, and maintain an immutable or otherwise tamper-evident audit trail. These controls shall enable full transparency and allow any reported emissions values or intensities to be independently verified and reproduced based on the underlying data and documented methodologies.

9 Data Quality Levels & Exception Reporting

9.1 Data Quality Levels

To incentivize continuous improvement in emissions monitoring and quantification processes while recognizing the practical challenges of real-world operations, facilities, and companies may progress along the following methodology levels as they acquire additional data. These current categorizations reflected below are *qualitative* at this time, though in the future it is intended to develop *quantitative* confidence intervals to indicate the uncertainty of the emissions reflected on the certificates.

- **Industry Emission Factor-Based:** emissions sources quantified using generic emissions factors from recognized industry publications, generally guided by 40 CFR 98 Subparts C & W and ONE Future. Analogous to OGMP 2.0 Level 3.
- **Measurement-Informed:** emissions sources quantified following Section 10 of this methodology, using source-level (bottom-up) calculations supported by site- and equipment-specific emissions and activity factors. These factors are derived from periodic measurements, including site surveys (e.g., LDAR) that capture equipment-level emissions, along with other approved measurement methods. Analogous to OGMP 2.0 Level 4.
- **Bottom-Up Top-Down Reconciled:** emissions sources quantified following this methodology, including periodic reconciliation of source-level (bottom-up) emissions with site-level (top-down) measurements in accordance with Sections 10 and 11. This includes site- and equipment-specific emissions and activity factors derived from source-level measurements (e.g., LDAR surveys and direct measurement), combined with top-down, site-level emissions reconciliation. Analogous to OGMP 2.0 Level 5.
- **Near CMS Bottom-Up Top-Down Reconciled:** emissions sources quantified following this methodology, with near-continuous monitoring system deployed onsite for Primary Emitting Facilities (defined in Section 10.1), allowing for rapid detection of abnormal conditions with potential for frequent automated top-down/ bottom-up reconciliation. This builds on and exceeds the requirements of OGMP 2.0 Level 5.

9.2 Exception Reporting

The categorizations mentioned above consider the materiality of different emissions sources for each facility. Meeting OGMP 2.0 Level 4 requires Level 4 measurements to be performed for sources that account for a target 90% of site emissions, with a minimum threshold of 70% with a justification for why 90% was not reached.

In some instances, companies or individual facilities may not have the ability to complete the detailed source-level monitoring outlined in Section 10 for certain material emissions sources. This may be due to the lack of available data or suitable measurement technology, or impractically high cost associated with measurement. Where measurements can be performed on sources that account for 70% of site emissions but cannot meet the 90% target, the sources without source-level measurement will be disclosed on the environmental attribute certificate for transparency. The emissions for these sources will be estimated using best-available data, which could include industry-wide or company-specific emission factors.

10 Bottom-Up Source-Level Emissions Calculations

The basic shared approach of GHG emissions calculation methods is to categorize all known types of GHG emissions and provide breakdowns by source type. The sources include continuous emission sources, as well as discontinuous or intermittent sources and events. With appropriate source-level measurements, routine methane emissions from many sources can be reasonably determined using engineering calculations based on periodic measurements combined with site

operating data. Bottom-up source-level calculations described in this section will be supported by top-down site-level measurements described in Section 11, which can identify the presence of unexpected or high-emitting sources not captured in the bottom-up calculations.

To satisfy measurement requirements defined by OGMP 2.0 and the GTI Veritas Protocol, source-level testing meeting OGMP 2.0 Level 4 is performed on sources expected to account for a target of >90% of site methane emissions (>70% minimum with acceptable written justification why >90% cannot be met—see Section 9.2). In order to ensure this target is met on an ongoing basis, at least annually the measurement results at each facility will be evaluated to determine whether new emissions sources have been identified and to assess whether a change in measurement technology or frequency is warranted to better capture and quantify site emissions.

10.1 Primary Emitting Facilities – Scope 1

Certain types of locations with more complex process operations and larger counts of potential emissions sources have been identified as primary emitting facilities. All primary emitting facilities of the following types will have rigorous measurement-informed bottom-up source-level emissions calculations described below, combined with top-down measurement technology described in Section 11:

- Wellpads
- Upstream Central Delivery & Processing Sites
- Compressor Stations
- Gas Treating Facilities
- Gas Processing Facilities
- NGL Fractionation & Storage Facilities
- LNG & Underground Storage Facilities
- Gas Transmission Facilities
- Natural Gas Fired Power Generation Facilities

Common scope 1 greenhouse gas emissions sources at natural gas facilities include:

- Combustion Engines and Turbines
- Reciprocating & Centrifugal Compressor Vents
- Reciprocating Engine Crank Case Venting
- Gas Starters
- Atmospheric Storage Tanks
- Gas-Driven Pneumatic Devices
- Desiccant Dehydrators
- Glycol Dehydrators
- Acid Gas Removal
- Well Venting for Liquids Unloading
- Casinghead or Annulus Vent
- Component Leaks
- Blowdowns
- Purges
- Control devices (e.g., flares, thermal oxidizers, and combustors)

Standard source-level calculations are enhanced through higher frequency measurement of unit or site-specific emissions rates. Bottom-up measurements will be conducted using the following methods:

For Upstream & Compressor Stations:

- Screening for leaks (Quarterly)
 - Screening could be performed using OGI surveys or Method 21 surveys
- Methane emissions mass rate measurement for certain sources (Quarterly)
 - Hi-Flow Sampler testing performed concurrently with LDAR surveys
 - Direct measurement of methane emissions mass rate for several sources:
 - Centrifugal compressor wet seal and dry seal gas vents
 - Reciprocating compressor rod packing seal vent
 - Blowdown vent through-valve leakage
 - Fugitive emissions detected during LDAR survey from equipment components, malfunctioning pneumatic devices, etc.
 - Storage tanks
- Exhaust gas stack testing (Annually)

For Gas Processing / NGL Fractionation Plants

- Screening for leaks
 - Complete annual survey. If results show that emissions from equipment leaks >5% of the overall site emissions, increase frequency and conduct semi-annual surveys for at least one year. If results show that emissions from equipment leaks >10% of the overall emissions, increase frequency and conduct quarterly surveys for at least one year.
 - Screening could be performed using OGI surveys or Method 21 surveys
- Methane emissions mass rate measurement for certain sources (concurrent with leak survey)
 - Hi-Flow Sampler performed concurrently with leak surveys
 - Direct measurement of methane emissions mass rate for several sources:
 - Centrifugal compressor wet seal and dry seal gas vents
 - Reciprocating compressor rod packing seal vent
 - Blowdown vent through-valve leakage
 - Fugitive emissions detected during LDAR survey from equipment components, malfunctioning pneumatic devices, etc.
- Exhaust gas stack testing (Annually)

For Natural Gas Fired Power Generation Facilities

- Exhaust gas measurement system or annual stack testing

Within this method, bottom-up source-level emissions estimates are calculated using well-established engineering relationship found in 40 CFR Part 98 Subparts C and W, and recognized industry methodologies, as detailed below and in Table 1. Until first quarter emissions measurements are made, historic facility measurements, company-based emissions factors, or industry-based generic emissions factors may be used as best available data.

1. Combustion Exhaust:

- Monitoring methodology:
 - Conduct annual stack test, with methane analysis, for each operating engine. For turbines and fired heaters, conduct stack test where feasible, otherwise use emission factors in Table W-7 to Subpart W of Part 98, Title 40.
 - Until first stack test is conducted, use stack test from same engine or turbine model at same facility; or
 - If stack test results are unavailable for the facility, use default methane emission factors in Table W-7 to Subpart W of Part 98, Title 40.
 - Measure monthly fuel use if possible, or calculate using run hours x heat rate
- CH₄ Calculation methodology:
 - Stack test result (lb CH₄/MMBtu) x mt/lb x fuel use (MMBtu/month).
- CO₂ Calculation methodology:
 - Stack test result (lb CO₂/MMBtu) x mt/lb x fuel use (MMBtu/month).
 - If stack test is unavailable, for pipeline quality fuel:
 - 40 CFR 98.33(a) Equation C-1, C-1a, or C-1b
 - If stack test is unavailable, for non-pipeline quality fuel:
 - Proposed 40 CFR Part 98.233(z) Equation-W-39A
- N₂O Calculation methodology:
 - For pipeline quality fuel
 - 40 CFR 98.33(c) Equation C-8 or C-8a
 - For non-pipeline quality fuel:
 - Proposed 40 CFR Part 98.233(z) Equation W-40

2. Compressor Vents:

- Monitoring methodology:
 - Conduct quarterly compressor monitoring survey. For each compressor:
 - Record as-found operating mode; and
 - Measure vented flow rates with Hi-Flow Sampler for:
 - Dry seal gas vents (pressurized mode); and
 - Wet seal degassing vents (pressurized mode); and
 - Rod packing vents (pressurized mode); and
 - Blowdown vents (if through-valve leakage is found during LDAR survey).
 - For compressors with dry seal gas vents, engineering estimate of leak rate may also be used where sufficient data is available
 - For each compressor, measure monthly operating hours in each of three operating modes:
 - Operating-pressurized mode; and
 - Standby-pressurized mode; and
 - Not operating-depressurized mode.
- CH₄ and CO₂ Calculation Methodology:
 - 40 CFR Part 98.233 (o)(6) Equations W-21, W-22, W-23 & W-24B
 - 40 CFR Part 98.233 (p)(6) Equations W-26, W-27, W-28, & W-29B
 - 40 CFR Part 98.233 (u) Equation W-35
 - 40 CFR Part 98.233 (v) Equation W-36

3. Reciprocating Engine Crankcase Venting:

- Monitoring methodology:
 - Track the number of crankcase vents per engine
 - Where feasible, conduct measurement of crankcase vent rate & composition
- CH₄ Calculation methodology:
 - If measurement is available
 - Measured vent rate x mol frac CH₄ x CH₄ density (mt CH₄/scf) x operating hours
 - If measurement is not available
 - Number of vents x 36 scfh CH₄ x CH₄ density (mt CH₄/scf) x operating hours
- CO₂ Calculation methodology:
 - Number of vents x Vent rate x mol frac CO₂ x CO₂ density (mt CO₂/scf) x operating hours

4. Engine Gas Starters:

- Monitoring methodology:
 - Track the number of engine starts monthly
 - Option 1: Measure fuel consumption for each start using fuel gas meter
 - Option 2: Determine fuel consumption per start from vendor documentation OR engineering estimate, and use calculations below
- CH₄ Calculation methodology:
 - # of starts x Fuel consumption/start x mol frac CH₄ x CH₄ density (mt CH₄/scf)
- CO₂ Calculation methodology:
 - # of starts x Fuel consumption/start x mol frac CO₂ x CO₂ density (mt/scf)

5. Atmosphere Storage Tanks:

- Monitoring methodology:
 - Monitor transmission storage tank vent with quarterly screening to detect upstream stuck-open dump valve(s);
 - Monitor gathering and processing storage tanks with regular screening or perform engineering estimate of vent emissions per 40 CFR Part 98.233 (j)
 - If continuous flow is detected, use Hi-Flow Sampler to measure leak rate.
- CH₄ and CO₂ Calculation methodology:
 - 40 CFR Part 98.233 (k)(1) – (k)(4) and 40 CFR Part 98.233 (j)(1)-(j)(7)

6. Gas -Driven Pneumatic Devices:

- Monitoring methodology:
 - Add gas-driven intermittent pneumatic devices to regular screening surveys for equipment components;
 - Use properly functioning EPA emission factor for properly functioning intermittent devices;
 - Use measured leak rate for malfunctioning devices where available, otherwise use malfunctioning intermittent devices EPA emission factor
 - Use EPA population emission factors for low-bleed and high-bleed continuous devices.
- CH₄ and CO₂ Calculation methodology:

- For intermittent devices:
 - 40 CFR Part 98.233 (a) Equation W-1B
 - 40 CFR Part 98.233 (a) Equation W-1C
- For low-bleed and high-bleed continuous devices:
 - 40 CFR Part 98.233 (a) Equation W-1B
 - 40 CFR Part 98.233 Table W-1
 - 40 CFR Part 98.233 (u) Equation W-35
 - 40 CFR Part 98.233 (v) Equation W-36

7. Desiccant Dehydrators

- Monitoring methodology:
 - Track the number of vessel openings monthly
 - For each vessel opening measure:
 - Gas pressure
 - Physical volume of desiccant dehydrator vessel
 - Void ratio of desiccant dehydrator vessel (i.e., % of packed vessel volume occupied by gas)
- CH₄ and CO₂ Calculation methodology:
 - 40 CFR Part 98.233 (e)(3) Equation W-6

8. Glycol Dehydrators

- Monitoring methodology:
 - Measure monthly operating hours; and
 - Measure annual representative inlet gas temperature, inlet gas pressure, inlet gas flowrate, and glycol circulation rate; and
 - Document glycol pump type (gas-driven or electric), use of flash tank separator, and controls on flash tank vent and regenerator vent.
- CH₄ and CO₂ Calculation methodology:
 - Conduct annual emissions modeling using GRI-Glycalc or Promax or other acceptable software to obtain lb CH₄/hr and lb CO₂/hr from flash tank vent, and lb CH₄/hr and lb CO₂/hr from regenerator vent.
 - 40 CFR Part 98.233 (e)(1)

9. Acid Gas Removal Units:

- Monitoring methodology:
 - Track the number of acid gas removal units per site
 - Conduct measurement of acid gas removal vent rate and concentration. Alternatively, perform engineering estimates using material balance or process simulation
- CH₄ Calculation methodology:
 - Proposed 40 CFR 98.233 (d) Calculation Methods 2-4
 - If no measurement or process simulation is available, use emission factor of 42762.8817 kg CH₄/ AGR / year for uncontrolled AGR vents.
- CO₂ Calculation methodology:
 - 40 CFR 98.233 (d) Calculation Methods 2-4

10. Well Liquid Unloading

- Monitoring methodology:

- Track the number of unloading events for each well monthly
- Record time that well was left open to atmosphere
- Method 1:
 - Measure vent gas flowrate by installing a recording flow meter on vent (measure wells with and without plunger lifts separately)
 - Performed annually, for at least one well per pad of each unique well tubing diameter group and pressure group (per 40 CFR 98.238).
- Method 2:
 - For wells without plunger lift:
 - Record shut-in pressure or surface pressure for wells with tubing production or casing pressure for wells without packers
 - For wells with plunger lift:
 - Record flow-line pressure based on engineering estimate
- CH₄ and CO₂ Calculation methodology:
 - Method 1:
 - 40 CFR Part 98.233 (f)(1) Equation W-7A and W-7B
 - Method 2:
 - 40 CFR Part 98.233 (f)(2) Equation W-8
 - 40 CFR Part 98.233 (f)(3) Equation W-9

11. Casinghead or Annulus Vent

- Monitoring methodology:
 - Record the number of monthly hours the annulus is vented to atmosphere
 - Measure and record annulus vent gas flowrate by installing flow meter on vent
- CH₄ Calculation methodology:
 - Vent rate (scf/hr) x mol frac CH₄ x CH₄ density (mt CH₄/scf) x vent hrs /month
- CO₂ Calculation methodology:
 - Vent rate (scf/hr) x mol frac CO₂ x CO₂ density (mt /scf) x vent hrs /month

12. Component Leaks (emissions from valves, PRVs, open-ended lines, flanges, threaded connectors)

- Monitoring methodology:
 - Conduct regular screening surveys to identify and repair leaks; and
 - Use Hi-Flow Sampler to measure leak rates in lieu of EPA emission factors. Hi-Flow Sampler device type should be known to inform if leak rate represents whole gas or methane only.
 - Leak duration will be calculated per the following approach:
 - If the start date of the leak is known, this date will be used. If the start date is unknown, the leak will be applied starting from the previous survey. The leak duration will end when it is confirmed to be repaired.
- Calculation Methodology for Devices that Measure Whole Gas Rate:
 - CH₄ Calculation methodology:
 - Leak rate (scf/hr) x mol frac CH₄ x CH₄ density (mt CH₄/scf) x leak hrs
 - CO₂ Calculation methodology:
 - Leak rate (scf/hr) x mol frac CO₂ x CO₂ density (mt /scf) x leak hrs
- Calculation Methodology for Devices that Measure Methane Rate:

- CH₄ Calculation methodology:
 - Leak rate (scf/hr) x CH₄ density (mt CH₄/scf) x leak hrs
- CO₂ Calculation methodology:
 - Leak rate (scf/hr) / mol frac CH₄ x mol frac CO₂ x CO₂ density (mt/scf) x leak hrs

13. Blowdowns at Primary Emitting Facilities:

- Monitoring methodology:
 - Track blowdowns monthly; and
 - For each blowdown, measure:
 - Gas temperature; and
 - Gas pressure at beginning of blowdown; and
 - Gas pressure at end of blowdown; and
 - Physical volume of equipment being blown down.
- CH₄ and CO₂ Calculation methodology:
 - 40 CFR Part 98.233 (i)(2) Equation W-14B OR Equation W-14A; and
 - 40 CFR Part 98.233 (u) Equation W-35; and
 - 40 CFR Part 98.233 (v) Equation W-36.

14. Control Devices (e.g., flares, thermal oxidizers, and combustors)

- Monitoring methodology:
 - Where a flare or thermal oxidizer is unlit, a DRE of 0% shall be used
 - Where methane slip from flares or combustors is expected to constitute >2% of site methane emissions OR there is reasonable concern of low methane destruction efficiency, conduct initial stack test or process simulation to determine DRE within three (3) months if technically feasible AND repeat annually thereafter. Scenarios include:
 - Expected low DRE from process design or stack/tip specification
 - Known mechanical damage to stack/tip
 - High methane emissions identified using a site-level flyover
 - Where methane slip from flares is expected to constitute <2% of site methane emissions AND there is no cause to suspect low methane destruction efficiency, a methane DRE of 98% may be assumed;
 - Measure waste gas volume to flare or thermal oxidizer monthly. In cases where measured data is not available, engineering estimates will be used in the interim with an acceptable written justification and documented exception.
- CH₄ Calculation methodology:
 - Gas vol (scf/month) x mol frac CH₄ x CH₄ density (mt CH₄/scf) x (1–DRE)
- CO₂ Calculation methodology:
 - 40 CFR Part 98.233(n) Equation W-20
 - 40 CFR Part 98.233 (u) Equation W-35
 - 40 CFR Part 233 (v) Equation W-36
- N₂O Calculation methodology:
 - Proposed 40 CFR Part 98.233(z) Equation W-40
 - 40 CFR Part 98.233 (u) Equation W-35
 - 40 CFR Part 233 (v) Equation W-36

15. Gas well venting during maintenance workovers (with and without hydraulic fracturing)

- Monitoring methodology
 - Track workovers monthly; and
 - For workovers with hydraulic fracturing only:
 - Track flowback duration and gas flow during:
 - a. (1) initial flowback to open pits/tanks (pre-separation), and
 - b. (2) flowback after gas is sufficient for separation.
 - Measure gas flow using vent-line or multiphase flow meters.
 - Record 30-day production rates and any injected gas volumes (if applicable).
- CH₄ and CO₂ calculation methodology
 - 40 CFR Part 98.233(h) Equation W-13A (without hydraulic fracturing)
 - 40 CFR Part 98.233(g) Equation W-10A OR W-10B (with hydraulic fracturing)
 - 40 CFR Part 98.233 (u) Equation W-35; and
 - 40 CFR Part 98.233 (v) Equation W-36.

10.2 Primary Emitting Facilities – Scope 2

In addition to Scope 1 emissions from primary emitting facilities, Scope 2 emissions should be included to provide a more comprehensive view of the overall emissions inventory. The Greenhouse Gas Protocol's Scope 2 Guidance¹² outlines the methodologies for quantifying these emissions.

The approach applied here uses the location-based method, which relies on grid-specific emission factors determined by the facility's geographic location. Where more granular data are available, such as contractual instruments or supplier-specific emission factors, the market-based method may be applied in place of grid-average emission factors, consistent with GHG Protocol guidance.

1. Electrical Usage:

- Monitoring methodology:
 - Track electricity usage per month
- CH₄ Calculation methodology:
 - Electricity usage (MWh) x Grid-specific emission factor (lb CH₄ / MWh)
- CO₂ Calculation methodology:
 - Electricity usage (MWh) x Grid-specific emission factor (lb CO₂ / MWh)
- N₂O Calculation methodology:
 - Electricity usage (MWh) x Grid-specific emission factor (lb N₂O / MWh)

10.3 Minor Emitting Facilities

Emissions that occur outside of Primary Emitting Facilities must also be accounted for. Typically, these locations contain fewer, less complex process operations and relatively small counts of potential emissions sources. As a result, emissions of significant volumes of greenhouse gases are not expected during normal operating conditions. Nonetheless, emissions of minor amounts of greenhouse gases can occur during normal operations, and significant emissions can occur during

abnormal conditions due to pipeline safety issues or system maintenance, such as blowing down a segment of the pipeline prior to performing a repair or upgrade.

- Pipeline Segments
- Valve Sites
- Meter Stations
- Pig Launchers/Receivers

A similar approach is taken combining bottom-up source-level emissions calculations described with top-down measurement technology described in Section 11. However, as these sources account for low portions of total system emissions and are distributed across large land areas, rigorous source-level measurement and near-continuous emissions monitoring is not always feasible. As a result, company-based emissions factors or industry-based emissions factors are relied on as best available data for bottom-up source-level emissions calculations.

1. Pipeline Leaks

- Monitoring methodology:
 - Obtain pipeline mileage by material type on a quarterly basis.
- Calculation methodology:
 - 40 CFR Part 98.233 (r) Equation W-32A; and
 - Use pipeline material type emission factor(s) in 40 CFR Part 98.233 Table W-1; and
 - 40 CFR Part 98.233 (u) Equation W-35; and
 - 40 CFR Part 98.233 (v) Equation W-36.

2. Pipeline Stations (including Valve Sites and Meter Stations)

- Monitoring methodology:
 - Obtain count of pipeline stations by station type
 - For Valve Sites, develop engineering estimates of population emissions factors for similar sites using best-available data to estimate the counts of individual equipment components at each site
- Calculation methodology:
 - 40 CFR 98.233 (r) Equation W-32A; or
 - 40 CFR 98.233 (r) Equation W-32B

3. Blowdowns of Pipeline Segments & Pig Launcher/Receivers

- Monitoring methodology:
 - Track blowdowns monthly; and
 - For each blowdown, measure:
 - Gas temperature; and
 - Gas pressure at beginning of blowdown; and
 - Gas pressure at end of blowdown; and
 - Physical volume of equipment or piping being blown down.
- Calculation methodology:
 - 40 CFR Part 98.233 (i)(2) Equation W-14B; and
 - 40 CFR Part 98.233 (u) Equation W-35; and
 - 40 CFR Part 98.233 (v) Equation W-36.

Table 1 - Bottom-up methane emissions monitoring and quantification summary

Methane Emissions Source	Calculation Methodology	Parameter Measured	Measurement Method / Instrument	Data & Measurement Frequency
Engine and Turbine Combustion Exhaust	Stack test result: (lb CH ₄ /Btu) x fuel use (Btu/month) - preferred;	Fuel Use; Run-hours	Measured Fuel (scf), Calculated Fuel (scf), Run-hours - SCADA or Ops Logs	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as available
	Stack test result (lb CH ₄ /hr) x mt/lb x hrs/month - alternate	lb CH ₄ / MMBtu of Fuel; lb CH ₄ / hr	Stack Test - EPA Approved Method	Annually
Centrifugal Compressor Vents	40 CFR Part 98.233(o)(6) Equation W-21 & W-22	Run hours in Operating Pressurized Mode	Run Status Indicator, Pressure Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
		Run hours in Standby Pressurized Mode	Run Status Indicator, Pressure Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
		Run hours in Not-Operating Depressurized Mode	Run Status Indicator, Pressure Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
	40 CFR Part 98.233(v) Equation W-36	As-found Mode	Visual Observation during Compressor Monitoring Survey	Quarterly
		Uncontrolled Dry Seal Gas Vent Measurement	Hi-Flow Sampler or Portable Flowmeter during Compressor Monitoring Survey, or Flowmeter-SCADA if present	Near-real time SCADA feeds for Flowmeter (primary); Quarterly for Surveys (secondary)
		Uncontrolled Wet Seal Gas Vent Measurement	Monitoring Survey, or Flowmeter-SCADA if present	Near-real time SCADA feeds for Flowmeter (primary); Quarterly for Surveys (secondary)
Reciprocating Compressor Vents	40 CFR Part 98.233(p)(6) Equation W-26 & W-27	Uncontrolled Blowdown Vent Measurement (through-valve leakage)	Hi-Flow Sampler or Portable Flowmeter during Compressor Monitoring Survey, or Flowmeter-SCADA if present	Near-real time SCADA feeds for Flowmeter (primary); Quarterly for Surveys (secondary)
		Run hours in Operating Pressurized Mode	Run Status Indicator, Pressure Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
		Run hours in Standby Pressurized Mode	Run Status Indicator, Pressure Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
	40 CFR Part 98.233(v) Equation W-36	Run hours in Not-Operating Depressurized Mode	Run Status Indicator, Pressure Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
		As-found Mode	Visual Observation during Compressor Monitoring Survey	Quarterly
		Uncontrolled Rod Packing Vent Measurement	Hi-Flow Sampler or Portable Flowmeter during Compressor Monitoring Survey, or Flowmeter-SCADA if present	Near-real time SCADA feeds for Flowmeter (primary); Quarterly for Surveys (secondary)
Reciprocating Engine Crank Case Venting	If measurement is available Measured vent rate x mol frac CH ₄ x CH ₄ density (mt CH ₄ /scf) x operating hours If measurement is not available Number of vents x 36 scfh CH ₄ x CH ₄ density (mt CH ₄ /scf) x operating hours	Vent rate and composition	Field surveys	Quarterly
Gas Starters	Number of starts x Fuel consumption per start x mol frac CH ₄ x CH ₄ density (mt CH ₄ /scf)	Number of starts, fuel consumption per start	Flowmeter if available; best available data and engineering calculations otherwise	Near-real time SCADA feeds (preferred); or monthly Engineering Calcs (alternate), as Available
Atmospheric Storage Tanks	40 CFR Part 98.233 (k)(1) - (k)(4) and 40 CFR Part 98.233 (j)(1) - (j)(7)	Non-Hydrocarbon Tanks (Gas Blowby from Stuck-open Dump Valves)	OGI Camera and Hi-Flow Sampler	Quarterly
		Hydrocarbon Liquid Tanks	Detailed Process Engineering Simulation	Monthly or Annually, based on Variability and Materiality
Gas-Driven Pneumatic Devices	For intermittent devices: Proposed 40 CFR Part 98.233 (a)(6) Equation W-18 (06/21/22)	Number of properly functioning pneumatic devices from quarterly OGI leak survey; and Number and leak rate of malfunctioning pneumatic devices from quarterly OGI leak survey	OGI Camera	Quarterly
	For low-bleed and high-bleed continuous devices: 40 CFR Part 98.233(a) Equation W-1; and 40 CFR Part 98.233 Table W-1A; and 40 CFR Part 98.233 Table W-38; and 40 CFR Part 98(u) Equation W-35; and 40 CFR Part 98(v) Equation W-36	Count of Continuous Low Bleed Devices; and Count of Continuous High Bleed Devices	Manual Count from Operations; Monthly MQCs	Monthly
Desiccant Dehydrators	40 CFR Part 98.233 (e)(3)	Vessel opening count, gas pressure	Pressure Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
Glycol Dehydrators	40 CFR Part 98.233(e)(1)	Glycol Pump Run Hours	Glycol pump run status or Inlet Gas Flowrate - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
		Dehy vent gas flow rate and composition	Detailed Process Engineering Simulation	Annually
Acid Gas Removal Units	Proposed 40 CFR 98.233 (d) Calculation Methods 2-4 If no measurement or process simulation is available, use emission factor of 42762.8817 kg CH ₄ / AGR / year for uncontrolled AGR vents.	Control on Flash Tank, Control on Regenerator Vent	Control Hours - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
		Glycol Pump Type, Use of Flash Tank Separator	Ops Log	Annually
Well Liquid Unloading	Method 1: o40 CFR Part 98.233 (f)(1) Equation W-7A and W-7B	Vent gas flow rate (Method 1)	Flow Meter	Near-real time SCADA feeds (preferred), or monthly Ops Logs for local flowmeter (alternate), as Available
	Method 2: o40 CFR Part 98.233 (f)(2) Equation W-8 o40 CFR Part 98.233 (f)(3) Equation W-9	Shut-in pressure or surface pressure for wells with tubing production or casing pressure for wells without packers (Method 2)	Field instrumentation	Near-real time SCADA feeds (preferred), or monthly Ops Logs for local meters (alternate), as Available
Casinghead or Annulus Vent	Vent rate (scf/hr) x mol frac CH ₄ x CH ₄ density (mt CH ₄ /scf) x vent hrs/month	Number of hours per month that venting occurred Annulus vent flow rate	Ops Log Flow Meter	Monthly Near-real time SCADA feeds (preferred), or monthly Ops Logs for local flowmeter (alternate), as Available
Equipment Components (Fugitive emissions from valves, PRVs, OELs, flanges, threaded connectors)	Leak rate (scf/hr) x mol frac CH ₄ x CH ₄ density (mt CH ₄ /scf) x hrs leaking / month	Number of leaking equipment components; and Leak rate (scf/hr) of each equipment component during quarterly leak survey	OGI Camera and Hi-Flow Sampler, Gas Chromatograph	Quarterly
		Number of hours per month that each equipment component leaked	Review of Leak Detection and Leak Repair Records in Leak Tracker Pro	Monthly
Blowdowns at Primary Emitting Facilities	40 CFR Part 98.233(j)(w) Equation W-14B; and 40 CFR Part 98.233(u) Equation W-35; and 40 CFR Part 98.233(v) Equation W-36	Blowdown Count, Beginning Pressure, Ending Pressure, Temperature, Physical Volume	Facilities: Pressure Indicator, Temperature Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available
Control Devices (e.g., flares, thermal oxidizers, and combustors)	Waste gas volume (scf/month) x mol frac CH ₄ x CH ₄ density (mt CH ₄ / scf) x (1 - Destruction & Removal Efficiency (DRE))	Destruction and Removal Efficiency (DRE) Run hours Waste Gas Throughput (scf)	No measurement - 98 % DRE Run Status Flowmeter if available; best available data and engineering calculations otherwise	Annual Near-real time SCADA feeds (preferred); or monthly Engineering Calcs (alternate), as Available
Electrical Usage (Scope 2)	Electricity usage (MWh) x Grid-specific emission factor (lb CH ₄ / MWh)	Monthly electrical usage	Company billing records	Monthly
Pipeline Leaks	40 CFR Part 98.233(r) Equation W-32A 40 CFR Part 98.233 Table W-1A 40 CFR Part 98.233 (u) Equation W-35 40 CFR Part 233 (v) Equation W-36	Pipeline mileage (mi) by Material Type	GIS Department Database	Quarterly
Pipeline Stations (including valve sites and meter stations)	40 CFR 98.233 (r) Equation W-32A; or 40 CFR 98.233 (r) Equation W-32B	Count of pipeline stations by type Valve sites - Engineering estimates of EF to estimate counts	GIS Department Database	Quarterly
Blowdowns of Pipeline Segments and Pig Launcher/Receivers	40 CFR Part 98.233(j)(2) Equation W-14B; and 40 CFR Part 98.233(u) Equation W-35; and 40 CFR Part 98.233(v) Equation W-36	Blowdown Count, Beginning Pressure, Ending Pressure, Temperature, Physical Volume	Pipelines: Pressure Indicator, Temperature Indicator - SCADA or Ops Log	Near-real time SCADA feeds (primary), or monthly Ops Logs (secondary), as Available

10.4 CO₂ Equivalence

To have a complete picture of the GHG emissions in a single metric, the equation and factors from IPCC global warming potential (GWP)¹³, the Environmental Defense fund (EDF)¹⁴, and EPA GHGRP¹⁶ in the table below should be utilized. Note that fluorinated gases are not process-related to the natural gas value chain and can therefore be excluded from carbon intensity calculations.

$$CO_2e (mt) = CO_2(mt) + X * CH_4(mt) + Y * N_2O(mt)$$

Where X = applicable CH₄ factor from tables below and Y = applicable N₂O factor from tables below.

Table 2 - GWP100 Factors

Component	Fourth Assessment Report (AR4)	Fifth Assessment Report (AR5)	Sixth Assessment Report (AR6)
CO ₂	1	1	1
Methane (CH ₄) – Fossil (Fugitives)	25	28	29.8
Methane (CH ₄) - Non-Fossil (combustion + remaining)			27
Nitrous oxide (N ₂ O)	298	265	273

Table 3 - GWP20 Factors

Component	Fourth Assessment Report (AR4)	Fifth Assessment Report (AR5)	Sixth Assessment Report (AR6)
CO ₂	1	1	1
Methane (CH ₄) – Fossil (Fugitives)	72	85	83
Methane (CH ₄) - Non-Fossil (combustion + remaining)		84	81
Nitrous oxide (N ₂ O)	289	264	273

10.5 Statistically Representative Sampling

Previous sections of this methodology specify measurement approaches to be completed in support of reliable, measurement-based emissions quantification. It is nonetheless recognized that performing these measurements on every individual emissions source or device may not be operationally or economically feasible, and may not produce an improved understanding of actual emissions profiles in all cases due to the law of large numbers. Where equipment or devices associated with emissions sources are sufficiently similar, statistical sampling may be utilized to enable the development of measurement-based equipment- or source-specific emissions factors.

Groupings of equipment may consider equipment class/type, operating mode, manufacturer/model, time in operation, etc. Generally, groupings of similar equipment should be assessed for populations following similar design bases, typically an asset/operating unit, division, or segment level rather than a single-site or whole-company level.

To be representative, a minimum number of independent emissions source measurements must be captured. The population size listed in **Error! Reference source not found.** below refers to the total population of individual devices or emissions sources of a given type. In general, measurement campaigns are likely to be coordinated and conducted on a site-by-site basis for cost and time efficiency. Enough sites should be included to ensure that the total number of emission sources sampled exceeds the sample size reflected in the table below as a percentage of the total population of devices. Note, statistical sampling under this methodology is applied to bottom-up test methods and the deployment of near-continuous emissions monitoring devices only. Top-down, site-level flyovers should be performed for all covered facilities to screen for large, unexpected sources not included in the measurement-informed bottom-up emissions quantification.

Site selection for measurement campaigns should consider the risk of large or abnormal emissions. Risk factors could include facility age, history of emissions or large emissions events, unstable process operations, thermal cycling from frequent shut-down & restart, high operating pressure, high throughput, different operational modes, etc. While there is no prescriptive approach to determining which sites/equipment should be sampled, best effort should be made so that all sites/equipment are surveyed over multiple measurement cycles, though higher-risk facilities should be revisited more frequently. All applicable site/facility types (e.g. well sites, compressor stations, gas processing facilities, liquefaction plants, etc.) should be represented within a sampling campaign.

Year 1

When initiating a sampling program, company-specific statistical properties, such as the measurement mean and variance for each emissions source, may not be known. As a result, the sample size requirements will differ in year one in order to provide an initial sample set to estimate the company-specific measurement mean and variance to use for further refinement. The table below is modeled from Matrix 1 in OGMP 2.0 Reconciliation and Uncertainty Technical Guidance Practice¹⁰ for high-materiality sources (noting that, following OGMP guidance, for source-level quantification, virtually all sources fall within ‘Simple’ category.)

Table 4 - Statistically Representative Sample Size Requirements (modelled from OGMP 2.0)

Population Size	Sample Size
< 10	>20%
> 10	>15%
> 100	>10%
> 1,000	>5%

Year 2+

In subsequent years, when more is known about company-specific measurement distributions, the sample size should be calculated using established statistical methods to achieve a desired uncertainty and confidence in emissions quantification. As one example, for a lognormal distribution, the required sample size can be calculated using the equation below referenced from GTI Veritas Guidance for Uncertainty Calculations (v2.0, Mar 2024)¹¹.

$$n = \frac{2s_y^2 + s_y^4 + 2d + \sqrt{4s_y^4 + 4s_y^2(s_y^4 - 2d) + (s_y^4 + 2d)^2}}{4d}$$

Where,

$$d = \left(\frac{\ln(b + 1)}{z_{1-\frac{\alpha}{2}}} \right)^2$$

s_y^2 = measurement sample variance

α = confidence level

b = uncertainty

At this time, reasonable ranges are permitted for confidence level ($\geq 90\%$) and uncertainty ($\leq 30\%$). Additional research is in progress to further refine and optimize statistically representative sampling and uncertainty in total emissions reporting to be included in future updates to this methodology.

11 Top-Down, Site-Level Measurements and Site Emissions Reconciliation

Several studies have demonstrated that a significant fraction of emissions originate from a relatively small fraction of sources with higher-than-expected emissions. The use of advanced top-down measurement technology within this methodology helps to more rapidly identify the presence of unexpected or high-emitting sources over and above those identified using bottom-up emissions covered in Section 10. This strategy will be implemented in order to perform timely remediation, to more accurately quantify resulting methane emissions, and to improve confidence in reported emissions values.

Top-down site-level monitoring is most commonly performed using aerial technologies to perform periodic surveys to measure whole-site methane emissions from planes or drones. Advanced methane detection satellite technologies could provide an alternative for top-down measurement with more frequent repass. At a minimum, periodic top-down surveys must be performed semi-annually at Primary Emitting Facilities and Minor Emitting Facilities.

Sensing technology is evolving rapidly and may provide additional tools to increase the frequency or accuracy of top-down measurement supporting reconciled quantification of full-site emissions. Advanced near-continuous emissions monitoring systems (near-CEMS) provide a powerful tool for low-latency emissions detection. Field trials are currently in progress to evaluate the effectiveness of using certain near-CEMS technologies with quantification capabilities as a high-frequency top-down measurement source—including tower-mounted lasers and hyper/multi-spectral gas cloud imaging (GCI) cameras with quantification capabilities.

- Monitoring methodology:
 - Semi-annual aerial (airplane) surveys will be conducted to:
 - Detect significant methane emissions sources and events not captured by bottom-up methods at facilities and along pipelines in order to expeditiously address them; and
 - Screen for temporal changes by facility and pipeline segment; and
 - Reconcile against methane emissions quantification from bottom-up methods.
- Quantification methodology:
 - Accumulated aerial scanning data will be processed through the surveyor's proprietary algorithms and will be analyzed by the surveyor's data technicians.
 - Methane concentration, mass, and plume dimension information will be calculated and provided by the aerial surveyor in a report. The vendor shall also provide uncertainty bands and detection thresholds for all flyover measurements or near-CEMS measurement devices.
- Reconciliation of top-down site-level measurements with bottom-up source-level calculations will be conducted after each aerial survey as follows:
 - First, determine and document if any portion of top-down measured emissions originate from outside the monitored facility and should be excluded from analysis.
 - Top-down measurement devices generally have higher minimum detection thresholds and may not capture all sources, particularly those with small emissions rates. Measurement devices also have uncertainty bounds that must be factored into the reconciliation analysis and may vary based on environmental conditions and distance. Detection threshold and uncertainty bound information will be determined for the top-down measurement based on vendor input and/or industry references.
 - Compare top-down measurement (including uncertainty range) with measurement-informed bottom-up inventory. Bottom-up inventory should be time-matched to the period when the top-down measurement was taken.
 - If total measurement-informed, temporally-matched bottom-up emissions are within range of the site-level measurement (including measurement uncertainty ranges), use bottom-up emissions calculations.
 - If total measurement-informed, temporally-matched bottom-up emissions are outside the range of site-level measurement (including measurement uncertainty ranges), a root cause analysis should be performed to try to determine the source of the differing emissions. The root cause analysis will incorporate knowledge of site operating conditions/ changes, engineering judgment, measurement uncertainty, and temporal effects.
 - Using facility operating and maintenance records, determine and document if site-level survey captures routine episodic emissions with high instantaneous emissions rate included in bottom-up emissions (e.g., compressor blowdown).
 - Using facility operating and maintenance records, determine and document if site-level survey captures routine episodic emissions with high instantaneous emissions rate NOT included in bottom-up emissions (e.g., depressurizing equipment for intrusive maintenance).
 - Calculate additional emissions either:

- Using starting pressure, ending pressure, temperature, and equipment volume as described in 10.1.13 Blowdowns; or
 - Using facility operating and maintenance data, estimate the start and end time of the elevated emissions to calculate:
$$[\text{Elevated Emissions Rate (lb/hr)} - \text{Normal Emissions Rate (lb/hr)}] \times \text{Duration of Event (hr)}$$
 - Evaluate whether emissions source is likely to reoccur and should be added to known sources calculated in Section 10.
- Determine if site-level survey captures emissions from a correctable defect
 - Once malfunction has been corrected, using facility operating and maintenance data or prior emissions testing, estimate the start and end time of the elevated emissions to calculate: $[\text{Elevated Emissions Rate (lb/hr)} - \text{Normal Emissions Rate (lb/hr)}] \times \text{Duration of Event (hr)}$
- If top-down measurement captures a new emissions source that is not included in bottom-up estimate, or measures a higher emissions rate for a known source, add the top-down measurement to the inventory.
- If significant uncertainty in the measurement estimates or source attributions is found to hinder root cause analysis, site-level measurement may be repeated to obtain a reliable sample to perform reconciliation or a change in the top-down measurement approach should be considered.

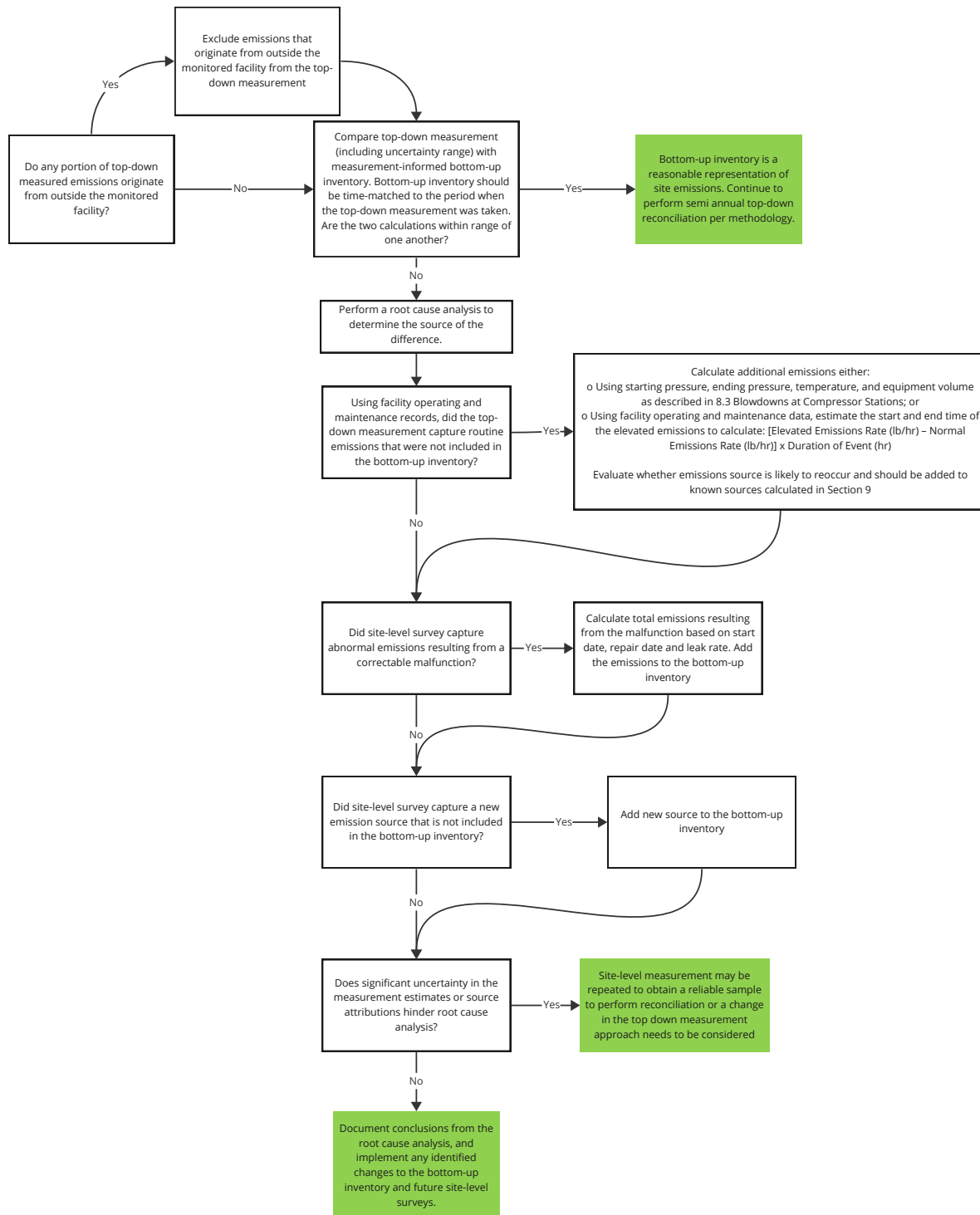


Figure 3. Reconciliation Process Flowchart

12 Emissions Allocation Method for End-to-End Emissions Tracking

For end-to-end emissions tracking from the wellhead to the final delivery point, the emissions associated with a specific oil or gas product shipment/delivery are added up for each emissions point across the supply chain. To accomplish this, the full flow path must be known from the point of origin (e.g., a well pad) to the final point of delivery (e.g., an industrial or utility offtaker, etc.).

Understanding emissions allocation at the well pad is relatively straightforward, as there is typically only one flow path out from the facility for the gas stream (wet or dry) and one for the liquid stream (if present). In this case, total methane emissions need only be allocated proportionately to the gas stream and liquid stream (if present). For the midstream, however, due to the nature of system operations, some additional analysis is required.

The sections below initially focus on conventional natural gas value chains. The scope is subsequently expanded to incorporate natural gas liquids (NGLs) to ensure comprehensive coverage of relevant hydrocarbon pathways.

The units of measure used to express emissions intensity in the following sections (e.g., % (mt/mt), kg CO₂e/MMBtu) reflect commonly applied industry conventions. Alternative units may be used where appropriate for a given pathway (e.g., kg CO₂e / kg propane), provided they are clearly defined and documented.

12.1 Production Segment

Where oil is produced simultaneously with gas, in accordance with the NGSI Methane Emissions Intensity Protocol V1.0 (2021), methane emissions should be allocated to each product stream proportional to the total energy produced in each stream. The following equations are reproduced from the NGSI Protocol:

The total energy produced (E_{ng}) in the gas stream is the product of the volume of gas produced (V_{ng}) multiplied by the energy content of the gas (EC_{ng}):

$$E_{ng} = V_{ng} * EC_{ng}$$

The total energy produced (E_{liq}) in the liquid stream is the product of the volume of liquids produced (V_{liq}) multiplied by the energy content of the hydrocarbon liquids (EC_{liq}):

$$E_{liq} = V_{liq} * EC_{liq}$$

The gas ratio is the total energy produced in the gas stream divided by the total energy of produced gas and liquids:

$$Gas\ Ratio\ (GR) = \frac{E_{ng}}{E_{ng} + E_{liq}}$$

The liquid ratio is the total energy produced in the liquid stream divided by the total energy of produced gas and liquids:

$$\text{Liquid Ratio (LR)} = \frac{E_{liq}}{E_{ng} + E_{liq}}$$

Component (CO₂, CH₄, and N₂O) emissions attributed to the gas stream and liquid streams are equal to the total component emissions at the site multiplied by the gas ratio and liquid ratio, respectively:

$$\text{Component Emissions}_{ng} = \text{Total Component Emissions} * GR$$

$$\text{Component Emissions}_{liq} = \text{Total Component Emissions} * LR$$

Where the total gas or total liquid produced by a single well pad is sold to different customers, the same emissions intensity per unit is applied to all units produced. Methane emissions per unit for gas or liquid are simply the ratio of the emissions allocated to a product stream to the total production for that same product stream. The emissions intensity of other components is calculated using a ratio of total emissions to total energy production for the same product stream, typically expressed in kilograms of emissions per MMBtu produced. This approach is appropriate because the majority of CO₂ and N₂O emissions are associated with combustion processes rather than direct gas releases.

$$\text{Methane Emissions per mt of Methane} = \frac{\text{Methane Emissions}_{ng}}{\text{Methane Production}}$$

$$\text{Methane Emissions per Bbl of Liquid Hydrocarbons} = \frac{\text{Methane Emissions}_{liq}}{\text{Liquids Production}}$$

$$\text{Component Emissions per MMBTU} = \frac{\text{Component Emissions}_{ng}}{\text{MMBTU Production}_{ng}}$$

$$\text{Component Emissions per MMBTU} = \frac{\text{Component Emissions}_{liq}}{\text{MMBTU Production}_{liq}}$$

12.2 Midstream Segment

Because midstream facilities handle flows from many customers at a time, further analysis is required. For the midstream segment, once the total emissions at each operating facility are known, the total emissions from each facility need to be equitably allocated to a specific oil or gas product shipment/delivery. That is, the total emissions are subdivided and allocated proportionately to a unique 'batch' of oil or gas represented by an environmental attribute certificate.

It is important to note that due to mixing and fungibility across the oil & gas supply chain, the exact molecule of natural gas received from the wellhead may not be the same molecule of natural gas physically delivered to the consumer. Based on pipeline operations, tracking molecules is not

possible. Nonetheless, a continuous flow path from a production source to a final delivery location with rights of ownership of the Environmental Attribute Certificates along the full path can be tracked relatively easily from end to end.

Across the midstream, emissions allocation occurs in four steps:

- (1) Construct a network graph of the end-to-end system
- (2) Identify the emitting facilities along the flow path
- (3) Determine total throughput and emissions per unit for each emitting facility
- (4) Determine total throughput and emissions per unit for pipeline segments

12.2.1 Construct a Network Graph of End-to-End System

The gas network consists of pipeline segments and locations. Locations are discrete points, and can be physical (e.g., a wellhead, a compressor station, etc.) or virtual (e.g., an administrative pooling location, etc., that exists only digitally with no physical counterpart). As an example, the Transco pipeline is comprised of over 1800 locations with pipeline segments connecting these locations. The position of a location can be identified using its pipeline milepost, i.e. the distance in miles from the origin of the pipeline.

By filtering the locations using the pipeline name and sorting them by milepost, a graph network of pipeline locations is generated by connecting each location to its adjacent neighbors.

Table 5 - Sample locations for Transco Mainline sorted for Milepost

Location ID	Short Name	Line Name	Milepost
9004065	WHARTON-TRES P M/L	TRANSCO MAINLINE	258.650
1003473	M4431 WH-VALERO XO	TRANSCO MAINLINE	265.910
1096690	GLEN FLORA (WHARTON)	TRANSCO MAINLINE	265.910
9006822	M3641 GLEN FLORA	TRANSCO MAINLINE	265.920
9009854	M4667 SPANISH CAMP XO	TRANSCO MAINLINE	271.200

Receipt locations are points where gas enters the pipeline network, while delivery locations are points where gas exits the pipeline network. Some locations (e.g., wellheads, etc.) act only as receipt points, while some other locations (e.g., power generation facilities, etc.) act only as delivery points. Some locations (e.g., physical storage facilities, pipeline interconnects, etc.) can act as either a receipt or a delivery point, while other locations (e.g., compressor stations, gas treating facilities, etc.) support pipeline operations and are neither a receipt nor a delivery location—instead, all gas in the pipeline passes through.

Certain types of locations with more complex process operations and larger counts of potential emissions sources have been identified as primary emitting facilities. All primary emitting facilities of the following types will have bottom-up source-level emissions estimates, combined with top-down measurement technology, following the procedure described in Sections 8 and 9:

- Wellpads
- Compressor Stations
- Gas Treating Facilities
- Gas Processing / NGL Fractionation Facilities
- Central Delivery Points
- Natural Gas Fired Power Generation Facilities

Note that primary emitting facilities are typically neither a receipt nor a delivery location. Note also that emissions between primary emitting facilities are not ignored, but rather measured or calculated and addressed as described in Section 10.2.

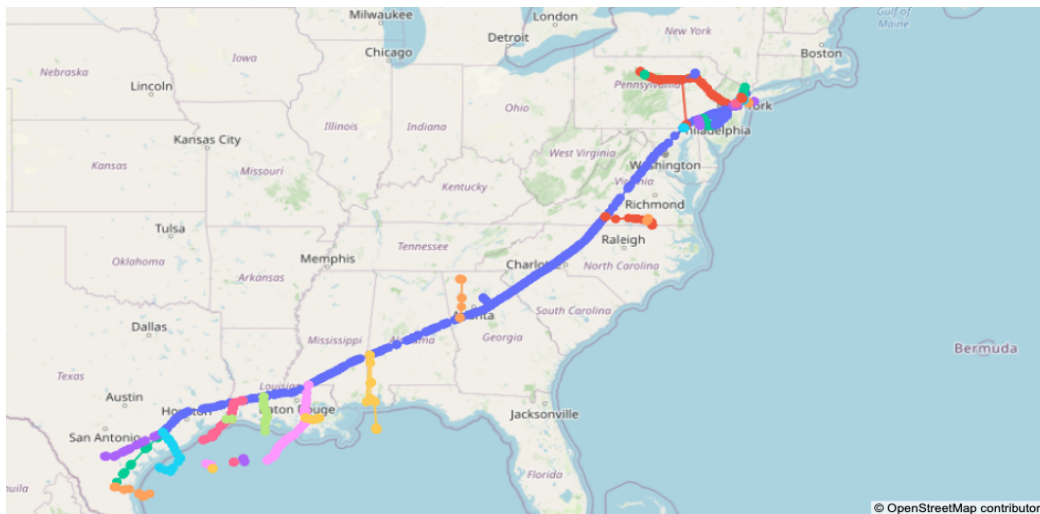


Figure 4. Map view of Transco pipeline network graph with >1800 locations connected by pipeline segment

12.2.2 Identify the Emitting Facilities Along Each Scheduled Flow Path

Once the network graph is constructed, after the flow day has occurred and allocations have been determined for each nomination, a list of primary emitting facilities along the flow path is tabulated for each nomination. First, the graph is traversed from the receipt point to the delivery point to identify the full list of locations along the flow path. Generally, it is assumed that the gas flows through the shortest path between the receipt location and the delivery location, unless a path route is specified (which could be driven by a nominated path, primary vs. secondary contract rights, etc.).

The full list of locations along the flow path is compared against the list of primary emitting facilities to determine the primary emitting facilities along the flow path.

12.2.3 Determine the Total Throughput and Emissions per Unit for Each Emitting Facility

Next, the total throughput at each emitting facility is determined and used to calculate the emissions per unit of throughput. Here, gathering & processing systems must be treated somewhat differently compared to transmission systems.

12.2.3.1 Gathering & Processing / NGL Fractionation

In a gathering system, oil or gas production enters from many well pads, but predominately flows in only one direction towards a relatively small number of interconnects into transmission pipelines. It is therefore straightforward to determine total throughput through a facility by directly observing the reading of a physical flow meter (or flow meters) recording the total flow through the facility.

Gathering & processing facilities may receive wet gas streams which are separated into a dry gas stream and a condensate or natural gas liquids (NGL) stream. For gathering & processing facilities where liquids are separated from gas, in accordance with the NGSI Methane Emissions Intensity Protocol V1.0 (2021), methane emissions should be allocated to each product stream proportional to the total energy produced in each stream. The following equations are reproduced from the NGSI Protocol:

The total energy produced (E_{ng}) in the gas stream is the product of the volume of gas produced (V_{ng}) multiplied by the energy content of the gas (EC_{ng}):

$$E_{ng} = V_{ng} * EC_{ng}$$

The total energy produced (E_{liq}) in the liquid stream is the product of the volume of liquids produced (V_{liq}) multiplied by the energy content of the hydrocarbon liquids (EC_{liq}):

$$E_{liq} = V_{liq} * EC_{liq}$$

The gas ratio is the total energy produced in the gas stream divided by the total energy of produced gas and liquids:

$$Gas\ Ratio\ (GR) = \frac{E_{ng}}{E_{ng} + E_{liq}}$$

The liquid ratio is the total energy produced in the liquid stream divided by the total energy of produced gas and liquids:

$$Liquid\ Ratio\ (LR) = \frac{E_{liq}}{E_{ng} + E_{liq}}$$

Component (CO_2 , CH_4 , and N_2O) emissions attributed to the gas stream and liquid streams are equal to the total component emissions at the site multiplied by the gas ratio and liquid ratio, respectively:

$$Component\ Emissions_{ng} = Total\ Component\ Emissions * GR$$

$$Component\ Emissions_{liq} = Total\ Component\ Emissions * LR$$

Methane emissions per unit for gas or liquid are simply the ratio of the emissions allocated to a product stream to the total throughput for that same product stream. The emissions intensity of other components is calculated using a ratio of total emissions to total energy throughput for the same product stream, typically expressed in kilograms of emissions per MMBtu of throughput. This

approach is appropriate because the majority of CO₂ and N₂O emissions are associated with combustion processes rather than direct gas releases.

$$\text{Methane Emissions per mt of Methane} = \frac{\text{Methane Emissions}_{ng}}{\text{Methane Received}}$$

$$\text{Methane Emissions per Bbl of Liquid Hydrocarbons} = \frac{\text{Methane Emissions}_{liq}}{\text{Liquids Received}}$$

$$\text{Component Emissions per MMBTU} = \frac{\text{Component Emissions}_{ng}}{\text{MMBTU Throughput}_{ng}}$$

$$\text{Component Emissions per MMBTU} = \frac{\text{Component Emissions}_{liq}}{\text{MMBTU Throughput}_{liq}}$$

12.2.3.2 Transmission

In a transmission system, gas can enter and exit the pipeline at many points along the pipeline. Commercial customers can nominate to ship gas in opposing directions at the same time. In physical operations, gas cannot flow bi-directionally within a single pipe at the same point in time. Instead, the pipeline operator optimizes their pipeline asset to move the least physical gas possible, while at the same time satisfying all commercial commitments. This results in a divergence between the total volume recorded by a physical meter at a compressor station along the line, and the total commercial commitments that a pipeline operator makes with a flow path across that point in the system. For this reason, the measure of throughput for the Transmission sector is defined as the total *received* volumes allocated for nominations with flow paths crossing a given facility.

This is best illustrated with an example. For a hypothetical segment of pipe on a given flow day, consider the scenario depicted below:

- Customer #1: Nominates 360,000 dth from milepost 5 to milepost 100 (South to North)
- Customer #2: Nominates 1,000,000 dth from milepost 65 to milepost 20 (North to South)

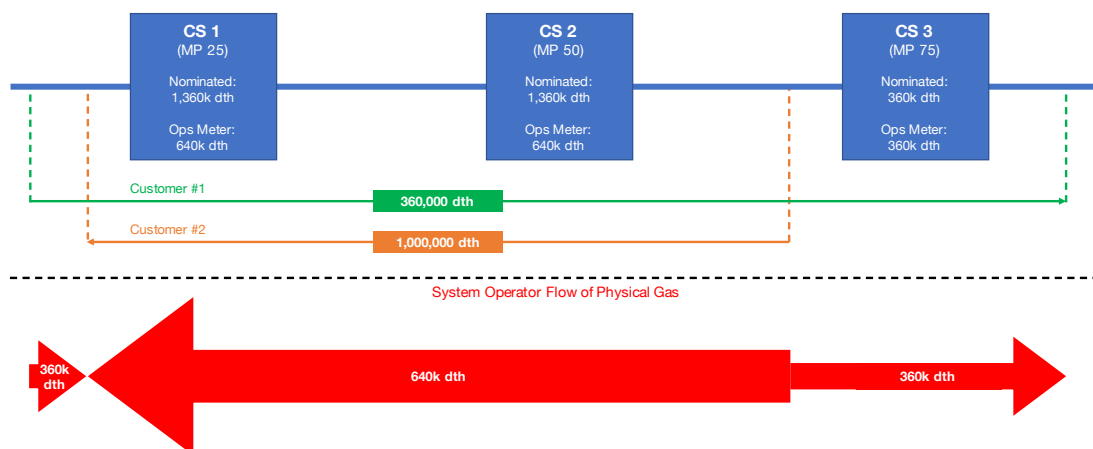


Figure 5. Commercial commitments vs. operational flow of physical gas

In this example, Customer #2 nominates to flow gas in the opposite direction compared to Customer #1. Physics will not allow flow in both directions at the same time, instead these opposing nominations result in *displacement*.

Assuming these are the only nominations on this day, in physical pipeline operations, Customer #1 will supply 360,000 dth at receipt point at milepost 5, which exits the pipeline at the delivery point for Customer #2 at milepost 20. Additionally, Customer #2 will supply 1,000,000 dth at the receipt point at milepost 65. This flow will be split, as shown, and delivered at milepost 20 (640k dth) and milepost 100 (360k dth). In this example, Customer #1 supplies 360,000 dth at milepost 5 and the same volume is delivered to the delivery point at milepost 100. Similarly, Customer #2 supplies 1,000,000 dth at milepost 65 and the same volume is delivered to the delivery point at milepost 20. However, an operational meter at Compressor Station #2 (milepost 50) will read only 640k dth, when the pipeline operator scheduled 1,360k dth of total commercial commitments to ship gas.

Due to the possibility of displacement described above, the measure of throughput for the Transmission sector is defined as the total *received volumes* allocated for nominations with flow paths across a given facility. (For illustrative purposes, using the example above, the throughput for Compressor Station #2 would be 1,360 dth.) This is done irrespective of whether a flow path is in the direction of net flow, or in the opposite direction. This method is preferred for several reasons:

1. Allows for proper overall mass balance for system emissions when all commercial shipments are certified
2. It more closely reflects the shipping choices made by customers using the pipeline system, rather than how the pipeline operator optimizes network flows
3. In the same way gas shipments can be scheduled in the direction of net flow and opposing net flow, it supports Certification both in the direction of net flow and opposing net flow
4. When taken system wide, it aligns with the NGS Protocol, which defines total system throughput as the *total volume of gas received*

Gas transmission pipelines process only ‘pipeline spec’ (i.e., dry gas). Therefore, there is no allocation required for different gas and liquid product streams. Once the total throughput per facility is known, methane emissions per mt of methane is simply the ratio of the facility methane emissions to the total methane throughput for that facility. The emissions intensity of other components is calculated using a ratio of total emissions to total energy throughput, typically expressed in kilograms of emissions per MMBtu of throughput. This approach is appropriate because the majority of CO₂ and N₂O emissions are associated with combustion processes rather than direct gas releases.

$$\text{Methane Emissions per mt of Methane} = \frac{\text{Methane Emissions}}{\text{Methane Received}}$$

$$\text{Component Emissions per MMBTU} = \frac{\text{Component Emissions}}{\text{MMBTU Received}}$$

12.2.4 Determine Emissions per Unit for Pipeline Segments

Finally, emissions that occur outside of primary emitting facilities must also be accounted for. These emissions can occur due to integrity related issues or system maintenance, such as blowing down a segment of the pipeline prior to performing a repair or upgrade. This can also include emissions associated with valve sites and meter stations. Activities like blowing down a pipeline segment affect large portions of a midstream asset and are necessary to support the system as a whole. For that reason, throughput is defined as total natural gas received within an operating Zone. The total amount of monthly emissions from pipeline segments, valve sites, and meter stations within an operating Zone will be allocated evenly to all units of monthly gas throughput within that same Zone.

$$(\text{Methane Emissions per mt of Methane})_{\text{Pipeline}} = \frac{(\text{Methane Emissions})_{\text{Zone}}}{(\text{Methane Received})_{\text{Zone}}}$$

$$(\text{Component Emissions per MMBTU})_{\text{Pipeline}} = \frac{(\text{Component Emissions})_{\text{Zone}}}{(\text{MMBTU Received})_{\text{Zone}}}$$

12.3 Natural Gas Fired Power Generation Facilities

For natural gas delivered to a power generation facility, emissions occurring at the facility can be categorized into two distinct components: (1) pre-combustion emissions and (2) combustion emissions during electricity generation.

Pre-combustion emissions at the facility include emissions associated with on-site fuel handling, compression, processing, or other activities that occur prior to fuel combustion. These emissions are incorporated within the pathway intensity described in this document.

Combustion emissions resulting directly from the generation of electricity are treated separately. In most commercial and reporting contexts, these emissions are accounted for independently and require a distinct intensity metric.

Accordingly, for the pathway intensity calculation referenced below, power generation facilities include only pre-combustion emissions occurring at the facility (i.e., emissions prior to fuel combustion for electricity generation). Emissions from fuel combustion during power generation are accounted for with a separate intensity from the natural gas pathway intensity. This distinction aligns with end-user reporting boundaries, where combustion emissions associated with purchased electricity are classified as Scope 2 emissions, while other emissions occurring at the facility may be treated differently depending on reporting context.

Pre-combustion emissions

$$\text{Methane Emissions per mt of Methane} = \frac{\text{Methane Emissions}_{ng}}{\text{Methane Inlet}}$$

$$\text{Component Emissions per MMBTU} = \frac{\text{Component Emissions}_{ng}}{\text{MMBTU Inlet}_{ng}}$$

Combustion emissions during electricity generation

$$\text{Component Emissions per MWh} = \frac{\text{Component Emissions}}{\text{MWh Delivered}}$$

12.4 Aggregate Emissions Across the Flow Path

Once the full flow path from the wellhead to the delivery point is established, emissions intensities are normalized to a common unit of throughput and aggregated across all segments of the pathway. Methane intensity, expressed as methane emissions per unit of methane throughput (e.g., mt CH₄ per mt CH₄ delivered), is summed across each emitting facility and pipeline segment along the flow path. Total methane emissions associated with a specific oil or gas delivery are calculated by multiplying the aggregated methane intensity by the total delivered mass.

An analogous approach is applied for carbon intensity, which is normalized to a common energy-based throughput unit across the pathway (e.g., kg CO₂e per MMBtu) and summed across all segments. Total carbon emissions associated with a given delivery are equal to the aggregated pathway carbon intensity multiplied by the delivered quantity expressed in the corresponding throughput unit.

Total End – to – End Methane Emissions per mt of Gas Certified:

$$\left(\frac{\text{Methane Emissions}}{\text{mt of Methane}} \right)_{\text{Total}} = \left[\sum_{\text{Emitting Facilities}} \left(\frac{\text{Methane Emissions}}{\text{mt of Methane}} \right) \right] + \left(\frac{\text{Methane Emissions}}{\text{mt of Methane}} \right)_{\text{Pipeline}}$$

$$(\text{Methane Emissions})_{\text{Total}} = \left(\frac{\text{Methane Emissions}}{\text{mt of Methane}} \right)_{\text{Total}} * \text{mt of Methane Certified}$$

Total End – to – End CO₂e Emissions per MMBTU of Gas Certified:

$$\left(\frac{\text{CO}_2\text{e Emissions}}{\text{MMBTU}} \right)_{\text{Total}} = \left[\sum_{\text{Emitting Facilities}} \left(\frac{\text{CO}_2\text{e Emissions}}{\text{MMBTU}} \right) \right] + \left(\frac{\text{CO}_2\text{e Emissions}}{\text{MMBTU}} \right)_{\text{Pipeline}}$$

$$(\text{CO}_2\text{e Emissions})_{\text{Total}} = \left(\frac{\text{CO}_2\text{e Emissions}}{\text{MMBTU}} \right)_{\text{Total}} * \text{MMBTU Certified}$$

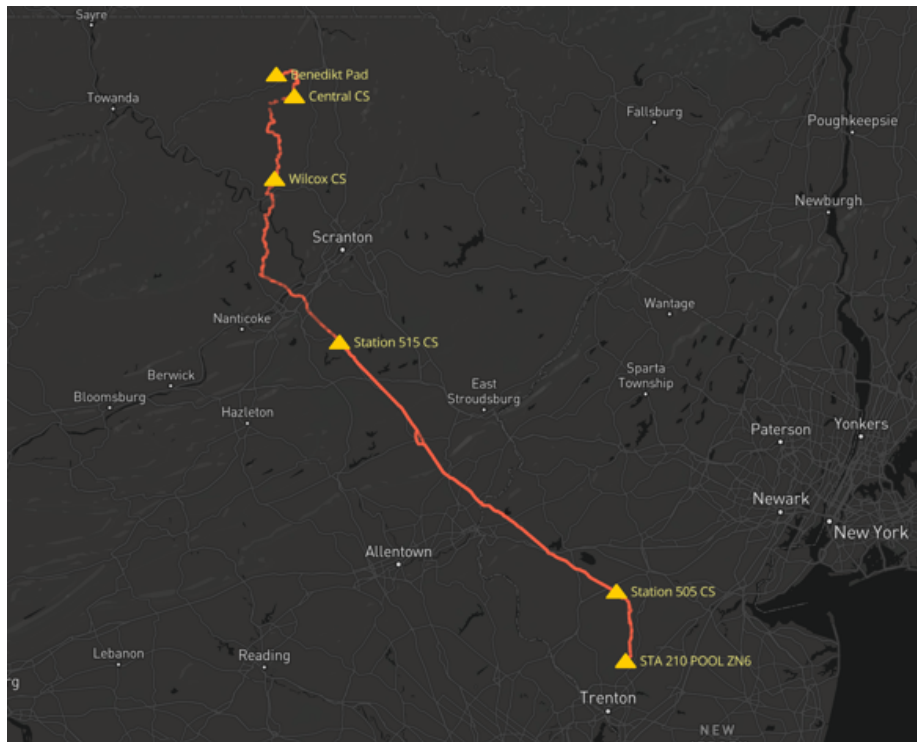


Figure 6. Emissions aggregation across the end-to-end flow path

12.5 Natural Gas Liquids

To incorporate Natural Gas Liquids (NGLs), emissions from mixed hydrocarbon streams (e.g., natural gas and associated NGLs) will be allocated using a physical energy-based allocation approach, consistent with PACT section 3.3.1.4¹⁵. Energy-based allocation is selected due to the strong underlying physical relationship between product streams and their respective energy content.

This approach is preferred over process subdivision, which is not likely to be feasible at scale across the industry, and economic allocation, which could introduce distortions unrelated to operational emissions drivers.

The energy allocation methodology aligns with the approach outlined in Sections 12.1 and 12.2.3.1 for attributing emissions between gas and liquid streams during production and gathering. However, for processing and fractionation facilities, the methodology will be expanded to allocate emissions across each individual value stream based on energy content, including natural gas, mixed NGLs, ethane, propane, butane, iso-butane, and pentanes+.

An illustrative example is provided below in which natural gas production proceeds through gathering, processing, and fractionation facilities to ultimately deliver liquid propane (NGL). The calculations presented are for demonstration purposes only. Actual conversion factors and volumes may vary, and the values shown are intended solely to illustrate the methodology.

In this example, the delivery requirement is 100,000 MMBtu per month of propane NGL. Table 6 demonstrates how this delivery requirement translates into total product quantities at each stage of the value chain.

Table 6 - NGLs example energy allocation

Facility Type	Product Stream	Quantity	UoM
Fractionation Outlet	Desired Propane delivered	100,000	MMBtu/month
		26,110	bbl/month
Fractionation Inlet	Required NGLs	262,860	MMBtu/month
Gas Processing Outlet		69,416	bbl/month
Gas Processing Inlet	Required Natural Gas	969,085	MMBtu/month
Production / Gathering outlet		944,528	Mscf/month

As illustrated in Table 7, however, a detailed quantity breakdown at each stage is not strictly required to determine the total pathway emissions intensity (although such analysis may be necessary to confirm that the selected pathway is operationally viable from a volumetric and mass balance perspective).

Under the energy allocation approach, once the emissions intensity is established for each facility along the pathway, the total pathway emissions are calculated by multiplying the applicable emissions intensity by the total energy content of the final delivered product. This approach allows total emissions associated with the relevant pathway to be determined without requiring stepwise volumetric scaling at each intermediate stage.

Table 7 - Example Pathway Intensities and Total Emissions

Segment	Propane Volume (MMBTU)	Overall Segment Intensity (kg CO ₂ e / MMBTU)	Overall Segment Intensity (kg CO ₂ e / kg C ₃ H ₈)	CO ₂ e Emissions (mt)
Production	100,000	0.5831	0.02792	58
Gathering	100,000	0.0297	0.00142	3
Processing	100,000	1.2610	0.06037	126
Fractionation	100,000	4.2218	0.20212	422
Transmission (Propane)	100,000	0.0227	0.00109	2
Total		6.1183	0.2929	612

13 Verification

A qualified third-party verifier will perform a comprehensive analysis to verify the integrity of data, its provenance, veracity, and calculations applied under this Methodology. The verifier will investigate the process by which emissions are generated, vet the approach used to create measurement-based emissions inputs and independently reperform aggregated emissions calculations for

verification purposes. The results of the verifier's independent calculation engine will allow for the preliminary screening of measurement-based emissions given historical data and relevant industry experience. With such context, baseline expectations will be set and outliers will be identified. Verifiers must be qualified to perform verification under this Methodology through a documented qualification process maintained by the Methodology owner, and must hold appropriate accreditation (typically ISO 14065, or the successor ISO 17029, accredited by an IAF MLA member body) for the scope of the engagement. The Methodology owner maintains a registry of qualified verifiers and may add additional verifiers as the Methodology is deployed in new contexts.

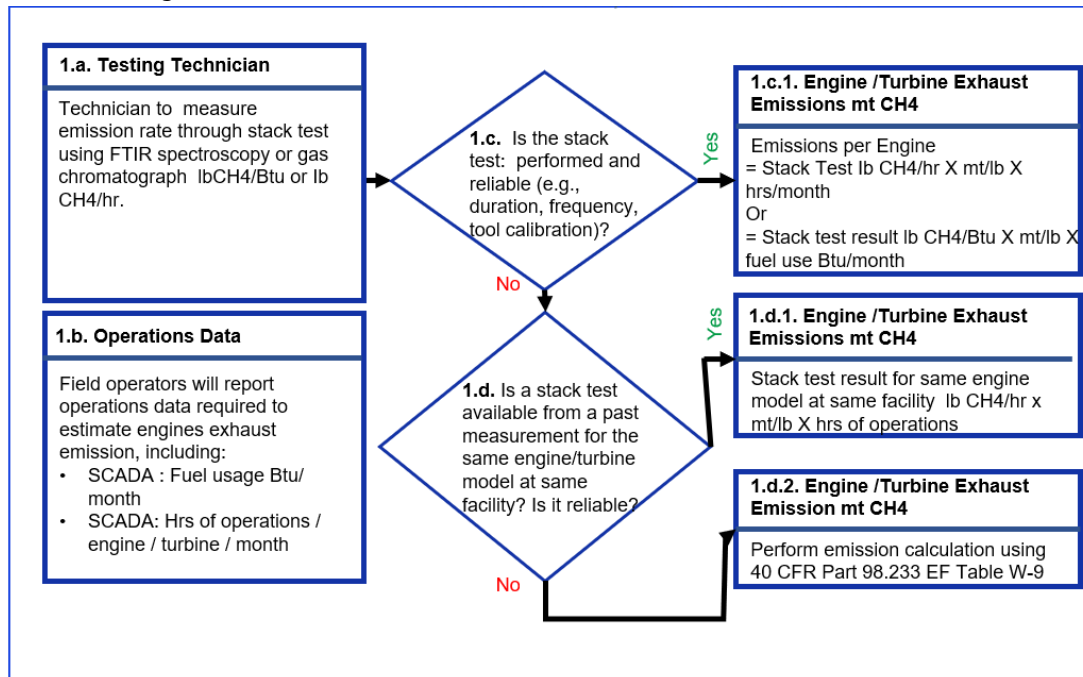
This release of the Methodology (v2.5) describes verification on a monthly batch basis, in which the verifier independently reviews each monthly batch of certificates produced from the application of this Methodology to a given operator's flow path. The Methodology owner is preparing a transition to a layered verification model in the next release (v2.6), under which platform-level methodology engine implementation verification will be performed once per Methodology version and applicant-specific verification will be conducted on an annual cycle with sample-based reperformance, defined material change triggers that activate interim verification, and explicit reliance on a SOC and ISO controls foundation maintained by the deploying carbon accounting platform. The annual model is intended to be backward-compatible with the batch model described in this section, such that verifiers and Methodology deployers operating under v2.5 will not be disadvantaged at the transition to v2.6. Until v2.6 is published, all Methodology applications continue under the monthly batch verification process described in this section.

To assess and verify aggregate emissions and intensities along the flow path, the verifier will focus on three areas: (1) understanding the approach used to create emissions inputs and site-wide emissions, (2) vetting measurement inputs, and (3) completing emissions calculations and associated screening and verification.

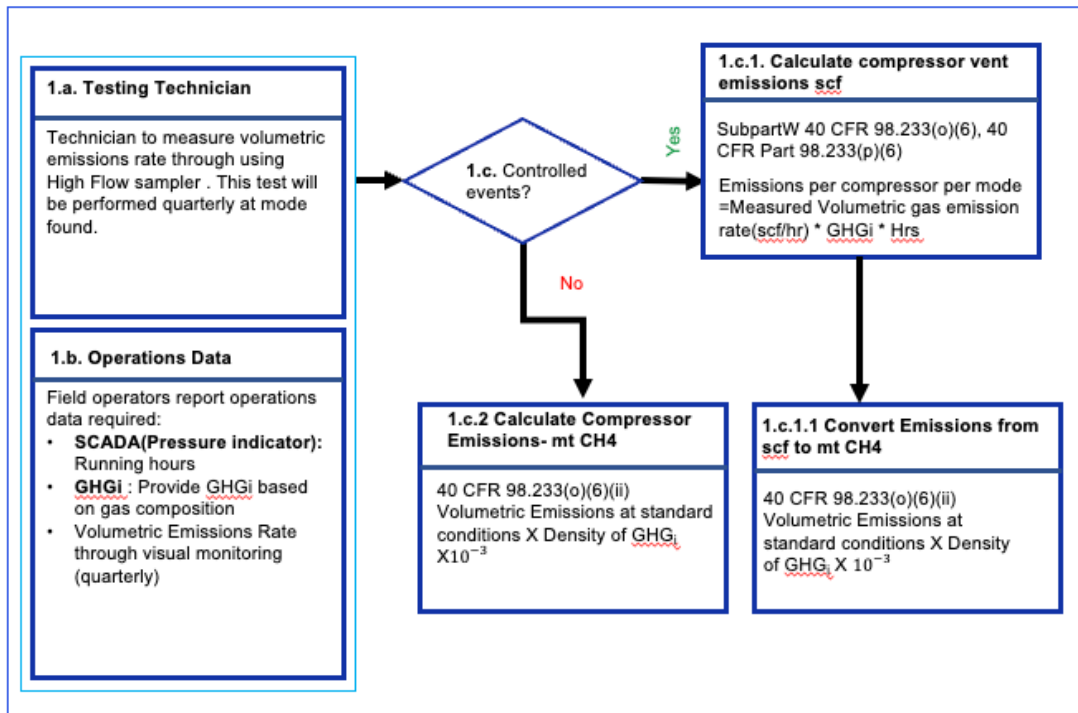
The facility operator's approach to generate higher-frequency measurement of unit- or site-specific emissions rates spans the following sources: (1) engine and turbine combustion exhaust, (2) reciprocating and centrifugal compressor vents, (3) gas starters, (4) reciprocating engine crank case venting, (5) atmospheric storage tanks, (6) gas-driven pneumatic devices, (7) desiccant dehydrators, (8) glycol dehydrators, (9) acid gas removal, (10) component leaks, (11) casinghead or annulus vent, (12) well liquid unloading, (13) blowdowns at emitting facilities, (14) control devices, (15) maintenance workovers, (16) electricity consumption, (17) pipeline leaks, and (18) pipeline blowdowns.

- 1) The process flows outlined below, for the eleven emission sources highlighted, document the steps the facility operator and the carbon accounting platform complete to translate on-site measurements and associated operational data into final emission values in metric tons.

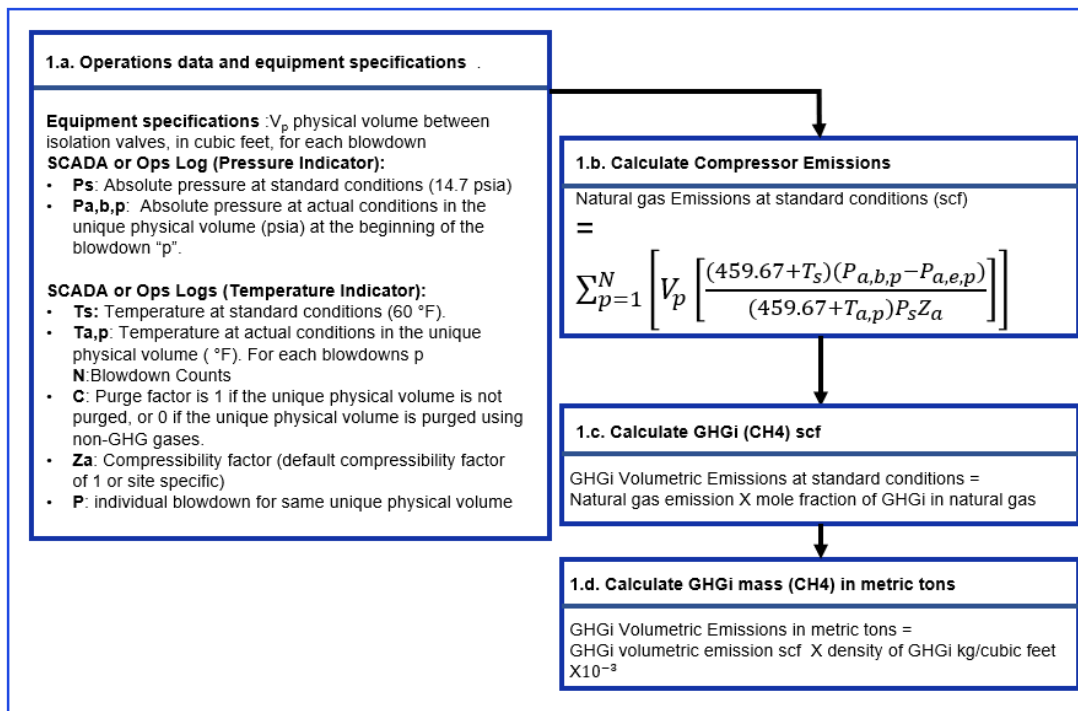
— 13.1.1.a. Engine and Turbine Combustion Exhaust



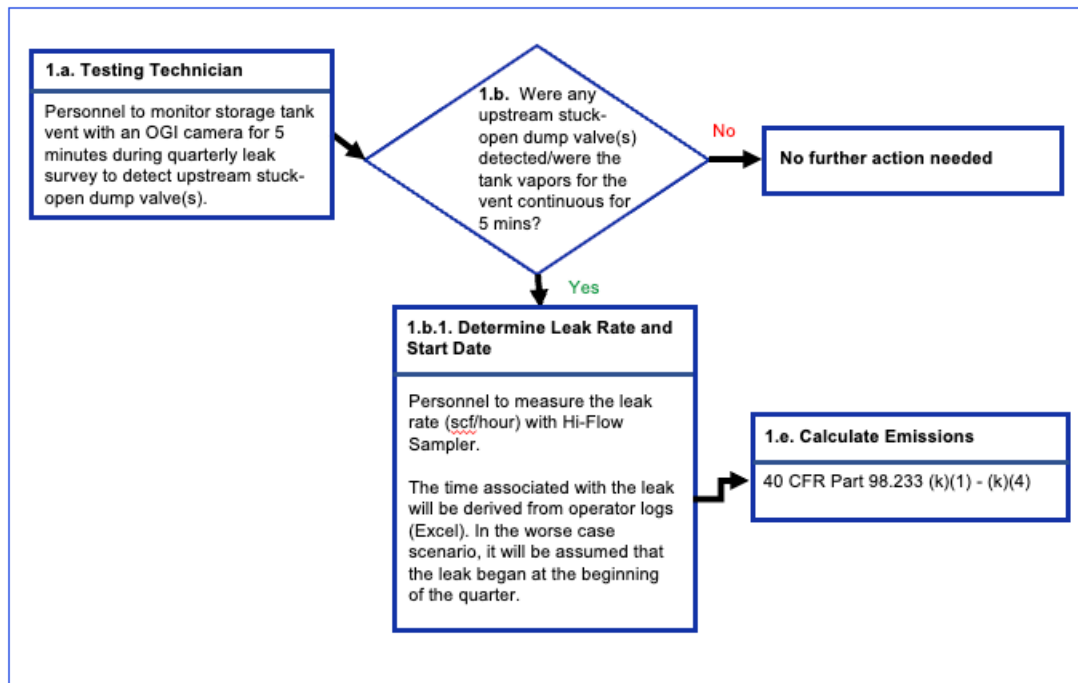
— 13.1.1.b. Compressor Vents



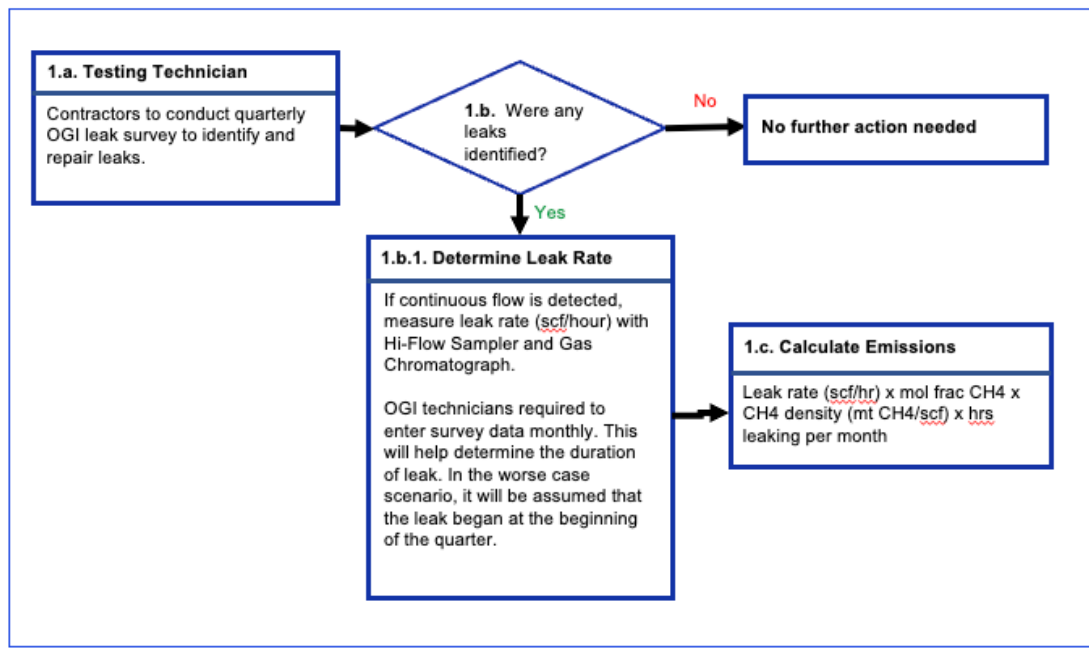
— 13.1.1.c. Blowdowns at Compressor Stations



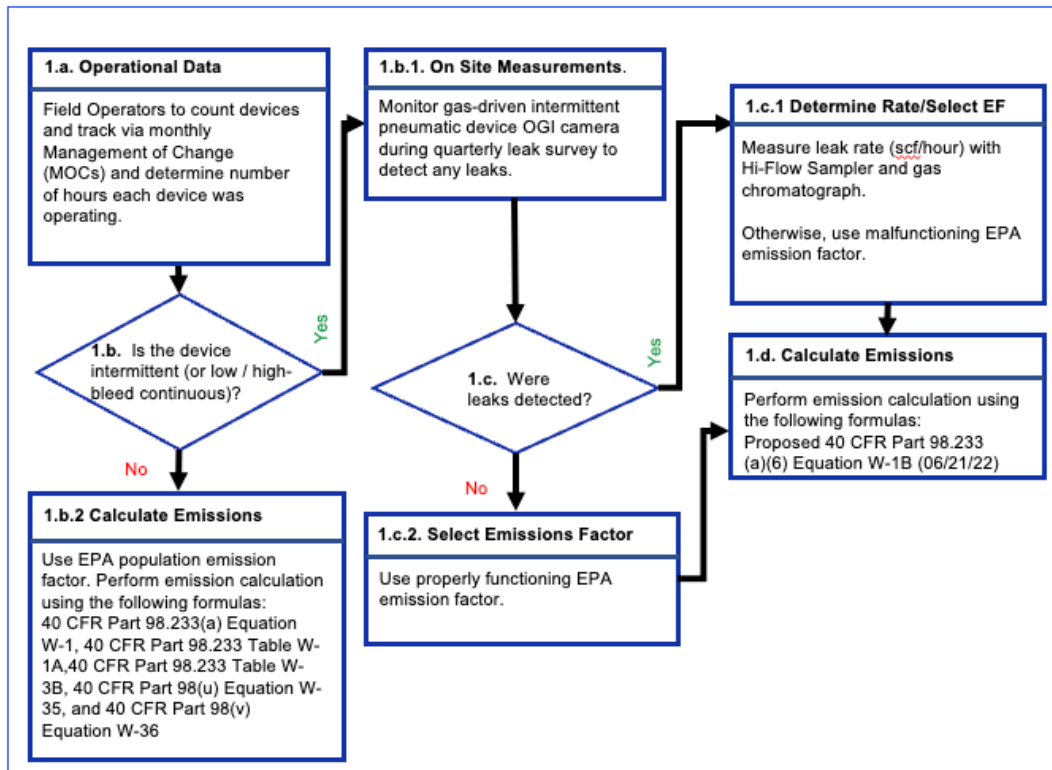
— 13.1.1.d. Atmospheric Storage Tanks (e.g., Stuck-Open Dump Valves)



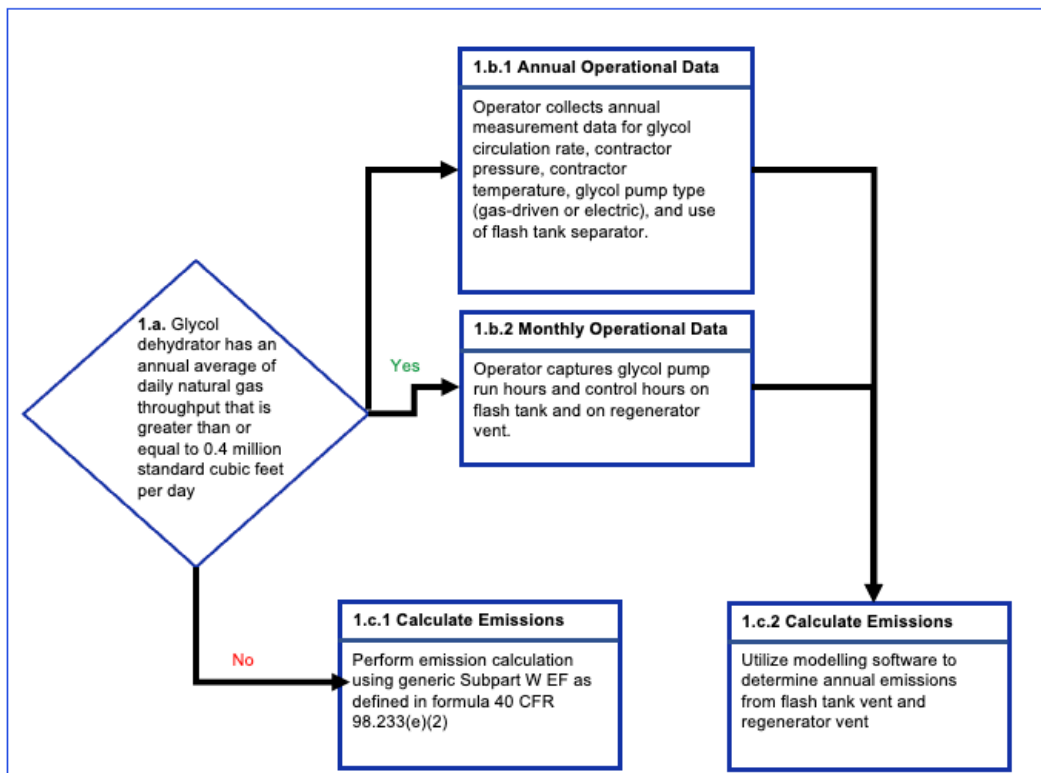
— 13.1.1.e. Equipment Components (Fugitive emissions from valves, PRVs, Open-ended lines, flanges, threaded connectors)



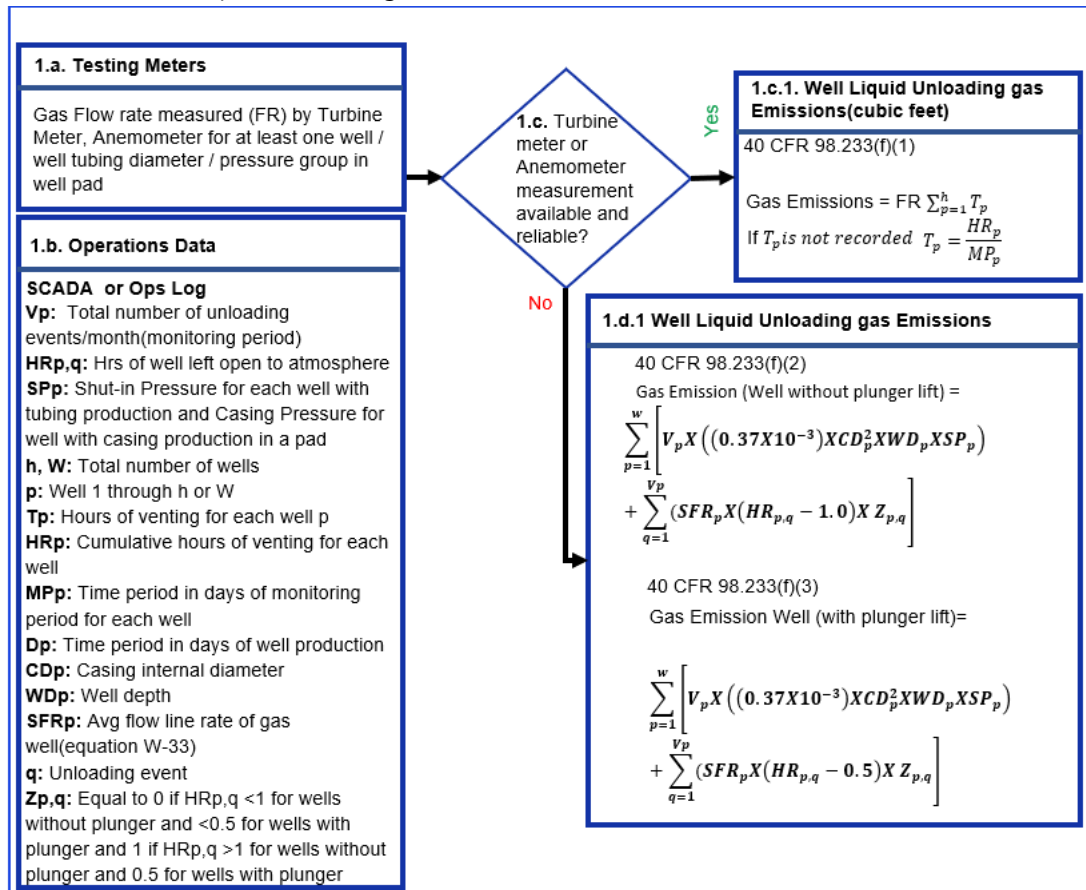
— 13.1.1.f. Gas-Driven Pneumatic Devices



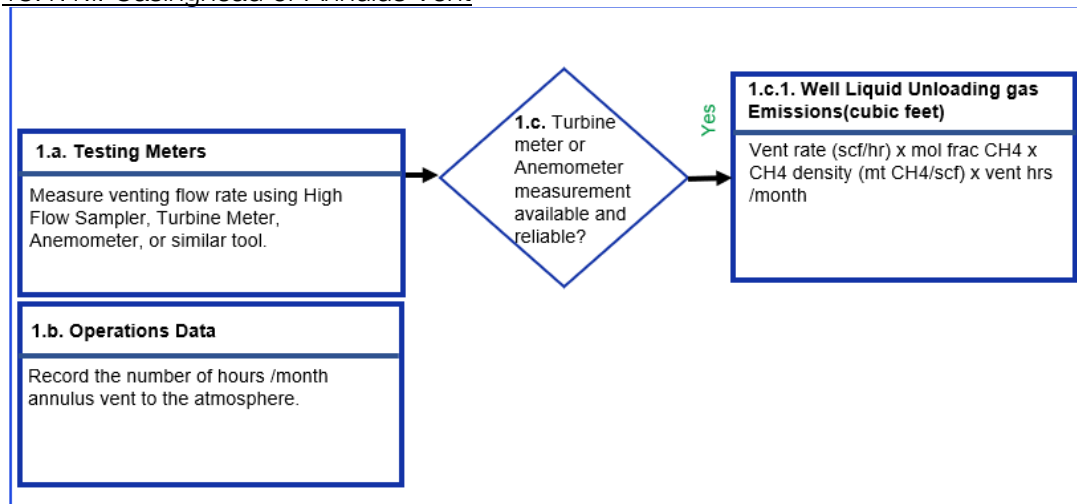
— 13.1.1.g. Glycol Dehydrators



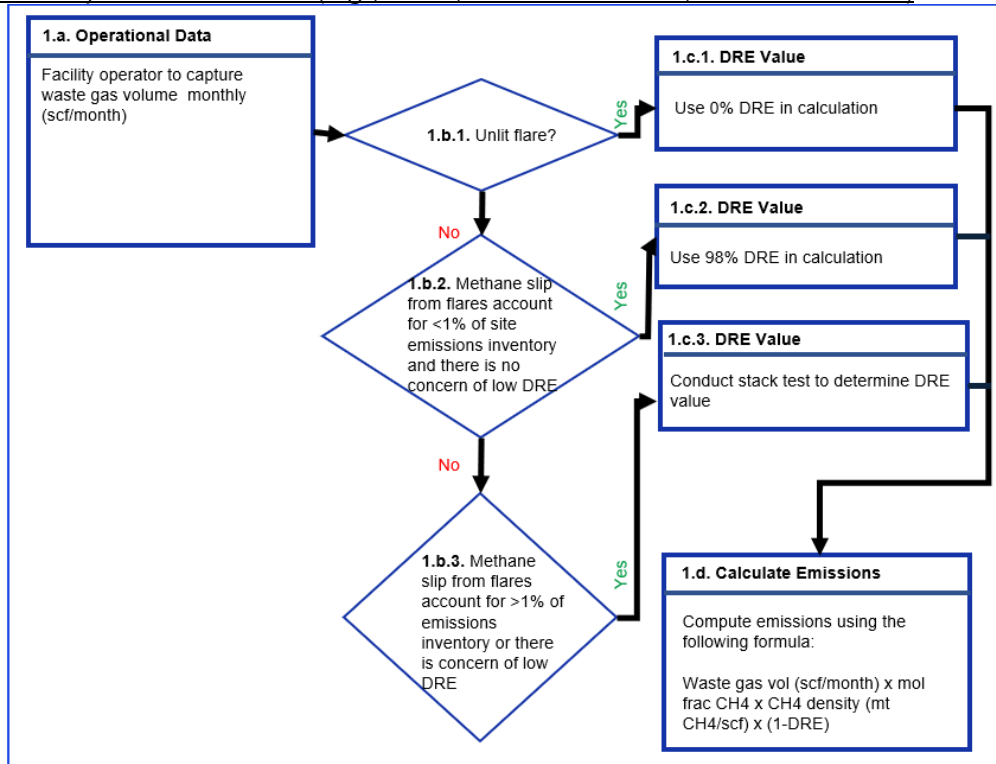
— 13.1.1.h. Well Liquid Unloading



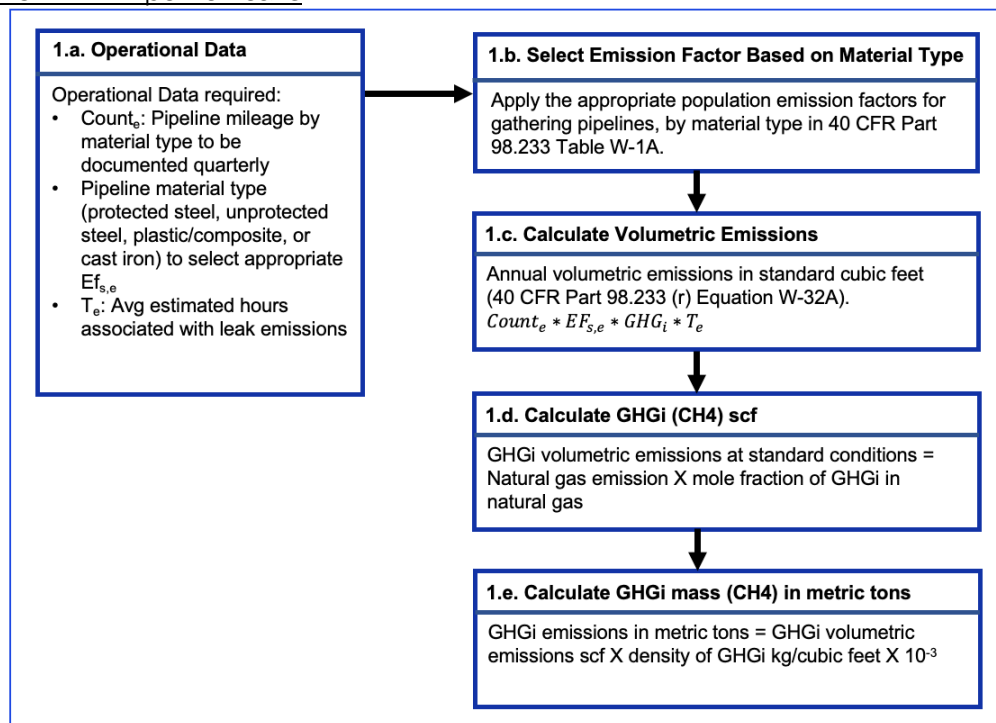
— 13.1.1.i. Casinghead or Annulus Vent



— 13.1.1.j. Control Devices (e.g., flares, thermal oxidizers, and combustors)



— 13.1.1.k. Pipeline Leaks



2) The verifier will vet the process of generating unit-specific or site-specific emission rates and values. The verifier will interview the carbon accounting platform's personnel and the facility operator's personnel to further understand the approach, as defined in Section 10 of this Methodology, to capture emission rates for all in-scope source categories. Responses will help inform the completeness and accuracy of sampling, measurement techniques, and calculation methods as depicted in the process flows for each source category in Section 13.1.1 of this Methodology. The verifier will utilize:

Original field measurement reports from accredited measurement providers and on-site source data to ensure data continuity from on-site measurements through to calculation inputs.

EPA literature, equipment manufacturer manuals, and industry best practices as guidance. Questions to be asked during this process can be found below.

Questions Applicable to Multiple Emission Source Categories

- How will emissions testing personnel/contractors mitigate Bacharach High-Flow Sampler limitations/deficiencies (e.g., failure modes investigation, the need for regular calibration)? (Applicable to: 13.1.1.b, 13.1.1.d, 13.1.1.e, 13.1.1.f, 13.1.1.i)
- How will emissions testing personnel/contractors ensure that the instrument is capturing all the gas escaping from the component(s)? (Applicable to: 13.1.1.d, 13.1.1.e)
- How will the leak start, and end dates be appropriately deduced from the operator logs? (Applicable to: 13.1.1.d, 13.1.1.e, 13.1.1.f)
- How do you ensure that the stack test procedures align with EPA guidelines and testing requirements (e.g., taking the test at constant temperature and constant pressure)? (Applicable to: 13.1.1.a, 13.1.1.j)
- Review the calibration frequency under operation conditions as the site operation conditions. (Applicable to: 13.1.1.h, 13.1.1.i)
- Do you measure the gas flow rate annually using the turbine meter? Or do measure it more frequently and take the average annually? (Applicable to: 13.1.1.h, 13.1.1.i)

Engine and Turbine Combustion Exhaust (13.1.1.a.)

- Confirm that the facility operator is completing stack tests for all turbine/engines at facilities
- How long are stack tests run for?
- If stack test results are not available for every engine, will facility operator use the test results from another device at the same facility and apply it accordingly? Will the facility operator use test results for the same engine model or leverage a historic test result for the same engine? How will difference in operating conditions be accounted for?

Compressor Vents (13.1.1.b.)

- How does the facility operator account for different operating modes (operating-pressurized mode; and standby-pressurized mode; and not operating-depressurized mode) during quarterly compressor monitoring surveys?

Equipment Components (13.1.1.e.)

- What is the process for an OGI leak survey (are all components monitored, how are the components monitored)?

Gas-Driven Pneumatic Devices (13.1.1.f.)

- Will all pneumatic devices in all sites be monitored on sites or will a sample be leveraged for extrapolation (or rather defaulted to industry standard EF)?
- Confirmation of whether EPA emission factors or on-site measurements will be taken for intermittent pneumatic devices.
- Why are high-bleed and low-bleed continuous devices not being monitored via on-site measurements?
- Will the number of hours each device was operating be tracked or will these numbers need to be estimated?

Glycol Dehydrators (13.1.1.g.)

- Is operational input data collected for all glycol dehydrators on site or just a sample?
- Is the previous year's data used to determine whether annual average of daily natural gas throughput is greater than or equal to .4 million scf / day?

Casinghead or Annulus Vent (13.1.1.i.)

- If the measurement is deemed unreliable, what actions will be taken (e.g., will an industry average be leveraged)?

Control Devices (e.g., flares, thermal oxidizers, and combustors) (13.1.1.j.)

- Is waste gas volume captured via flowmeter or engineering calculations?
- How does facility operator determine if methane slip from flares account for <1% of site emissions inventory and there is no concern of low DRE?
- For thermal oxidizers that require stack testing, are stack tests conducted annually?

Pipeline Leaks (13.1.1.k.)

- How will the average estimated hours associated with leaks be deduced (measured or a default applied)?

3) The verifier's independent calculation engine will ingest, process, and validate measurement data. The engine outputs — calculated Scope 1 methane emissions and average methane intensity for monthly batches of delivered gas — will be leveraged for the preliminary screening of the outputs and identification of outliers. As such, the verifier will:

- Flag monthly data quality as well as submittal issues and collaborate with the Methodology deployer to yield accurate calculation inputs.

- Verify executed calculations to ensure the translation of input measurements and operational data (e.g., equipment runtime, number of equipment) yields accurate intensities and aggregate emissions.
- Verify equipment-level emissions calculation integrity by spot-checking alignment with EPA Subpart W formulas and Section 10 of this Methodology.
- Calculate aggregate emissions and intensities along flow path segments and the entire flow path to confirm the platform's certificate output is in alignment with the Veritas Protocol or equivalent governing protocol applicable to the deployment.
- Preliminarily screen processes to form baseline expectations by (i) analyzing historical Subpart W emission factor-based baseline values for prior comparable periods and (ii) leveraging relevant industry expertise.
- Investigate anomalies in the data to understand potential causes (for example, to justify higher than historically observed emissions from specific equipment, or lower than expected throughputs or methane content) and discuss with the facility operator's engineering and/or environmental personnel as needed.

The verification approach described in this Section will evolve with the Methodology in v2.6 to support a more scalable verification model across multiple concurrent Methodology deployments. The v2.6 release will define the platform-level methodology engine implementation verification scope, the per-applicant annual verification scope and sampling methodology, the material change triggers that activate interim verification work, the reliance framework between verification tiers, and the qualification protocol for verifiers seeking to be added to the Methodology owner's registry. Methodology deployers and verifiers who require additional preparation guidance ahead of the v2.6 transition are encouraged to engage the Methodology owner directly.

14 References

1. Improving Characterization of Anthropogenic Methane Emissions in the United States. Washington, D.C., National Academies Press, 25 July 2018, www.ncbi.nlm.nih.gov/books/NBK519291/. Accessed 20 Nov. 2022.
2. Allen, David T., et al. "Consistent Metrics Needed for Quantifying Methane Emissions from Upstream Oil and Gas Operations." *Environmental Science & Technology Letters*, vol. 8, no. 4, 2021, pp. 345–349., <https://doi.org/10.1021/acs.estlett.0c00907>.
3. "NGSI Methane Emissions Intensity Protocol." American Gas Association (AGA) -- Natural Gas Sustainability Initiative, Feb. 2021, https://www.aga.org/contentassets/c87fc10961fe453fb35114e7d908934f/ngsi_methaneintensityprotocol_v1.0_feb2021.pdf.
4. Chollet, François. *Deep Learning with Python*. Manning, 2021.
5. Christ, Maximilian, and Nils Braun. "Tsfresh Documentation." <https://tsfresh.readthedocs.io/en/latest/>.
6. Pradhan, Manaranjan, and U. Dinesh Kumar. *Machine Learning with Python*. Wiley, 2019.
7. "Supervised Learning Models." Scikit Learn, https://scikit-learn.org/stable/supervised_learning.html.
8. Team, Keras. "Keras Documentation: Keras API Reference." Keras, <https://keras.io/api/>.
9. "Tensorflow V2.9.1 API Documentation." TensorFlow, https://www.tensorflow.org/api_docs/python/tf.
10. Guidance document: Reconciliation and Uncertainty (U&R) in methane emissions estimates for OGMP2.0. OGMP, 28 March 2022, <https://ogmpartnership.com/wp-content/uploads/2023/02/OGMP-2.0-UR-Guidance-document-SG-approved.pdf>. Accessed 5 July 2024
11. "Guidance for Uncertainty Calculations for Veritas Reporting." Veritas, <https://veritas.gti.energy/>
12. Sotos, Mary, et al. *GHG Protocol Scope 2 Guidance: An Amendment to the GHG Protocol Corporate Standard*. Greenhouse Gas Protocol, Mar. 2023, ghgprotocol.org/sites/default/files/2023-03/Scope%20%20Guidance.pdf.
13. Greenhouse Gas Protocol. Global Warming Potential Values. Aug. 2024, <https://ghgprotocol.org/sites/default/files/2024-08/Global-Warming-Potential-Values%20%28August%202024%29.pdf>.
14. Environmental Defense Fund. Emission Equivalency Tool: Documentation and Methodology. 23 June 2022,

https://www.edf.org/sites/default/files/content/emission_equivalency_tool_documentation_methodology_23062022.pdf.

15. World Business Council for Sustainable Development. PACT Methodology for Calculating and Exchanging Cradle-to-Gate Product Carbon Footprints (PCFs). Version 3.0, Mar. 2025.
16. "Table A-1 to Subpart A of Part 98." Electronic Code of Federal Regulations, Title 40, Chapter I, Subchapter C, Part 98, U.S. Government Publishing Office,
<https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-98/subpart-A/appendix-Table%20A-1%20to%20Subpart%20A%20of%20Part%2098>

Appendix A – About the Contributing Organization

Context Labs is an enterprise technology company dedicated to sourcing, organizing, contextualizing, and securing the world's climate information. The company enables trust in data, which can be utilized to inform markets. We synthesize disparate, disconnected data from all available data sources — including satellites, aerial flyovers, ground sensors, and other operational assets — to establish a more holistic understanding of ground truth carbon intensity for industrial operations. Data is ingested, contextualized, interconnected, and secured immutably using a combination of advanced graph analytics, artificial intelligence, and blockchain technologies—making climate disclosures fully auditable to prevent misrepresentation.

Recognizing the critical importance of eliminating anthropogenic methane emissions, this quantification methodology was developed using several years of R&D and has been tested in the market. This trusted emissions data supports the production of trusted environmental attributes with provenance direct to the source. Our near-real time emissions quantification is being used by leaders in the natural gas industry to support their decarbonization efforts. Context Labs' intent is to inspire the development of other emissions quantification methodologies that could also be applied using their AI-enabled data fabric platform for at-scale decarbonization of industrial operations.

Asset-Grade Data

Enhancing trust in data is at the heart of Context Labs technology. All data used to measure, model, or report on emissions or emissions intensity is supported by data surety processes to be considered "Asset-Grade Data" ("AGD"). Asset-grade data is data that has an immutable, auditable, end-to-end lineage to irrefutably prove the data has never been manipulated or otherwise corrupted. AGD is validated through the reconciliation of multiple sources, with accuracy attested by recognized third-party verifiers or subject matter experts.

Context Labs accomplishes this using several advanced methods.

- Contextual Data Fabric Technology: integrating and reconciling operational (SCADA) data and emissions data from multiple sources, including flyover, satellite, near-continuous emissions monitoring systems, and periodic leak and vent surveys (as available) to produce a more accurate representation of ground truth
- 'Digital Notaries': incorporating recognized third-party verifiers and subject matter experts that submit digitally signed statements attesting to the origin, integrity, and/or accuracy of a raw data input or analytics results
- Immutable Provenance: at each point where data is created, transformed, combined, analyzed, or accessed, a 'Digital Proof' is created and published to a distributed ledger (i.e., blockchain) for tracking and to immutably preserve data provenance and lineage throughout the full data lifecycle

Contextual Data Fabric

A contextual data fabric (CDF) is a centrally managed means for integrating disparate data sources intelligently and securely as a unified semantic data layer. The software architecture consists of a set of data services that provide consistent means to secure and access data integrated from a variety of discrete sources across multiple cloud environments. Advantages include:

- Data discovery and metadata-driven insights
- Data access and control
- Data privacy and security (protection)
- Data veracity (provenance and 3rd party attestations See Fig. 3).

A CDF is a data fabric system that uses extensive metadata to contextualize each data element. This metadata includes the pedigree and provenance (who, what, where, when) of the underlying data payload as well as contributions from other (often 3rd party) sources adding related information (facts) or attesting to the veracity of the underlying data. These additional facts (truths) can be used to analyze patterns of behavior, produce advanced analytics, or quantify the degree to which data can be trusted. At the heart of these capabilities lies graph analytics, which is used to infer relationships (connect data to other data) and structure (understand the relatedness or interdependencies between data, allowing the synthesis of new data).

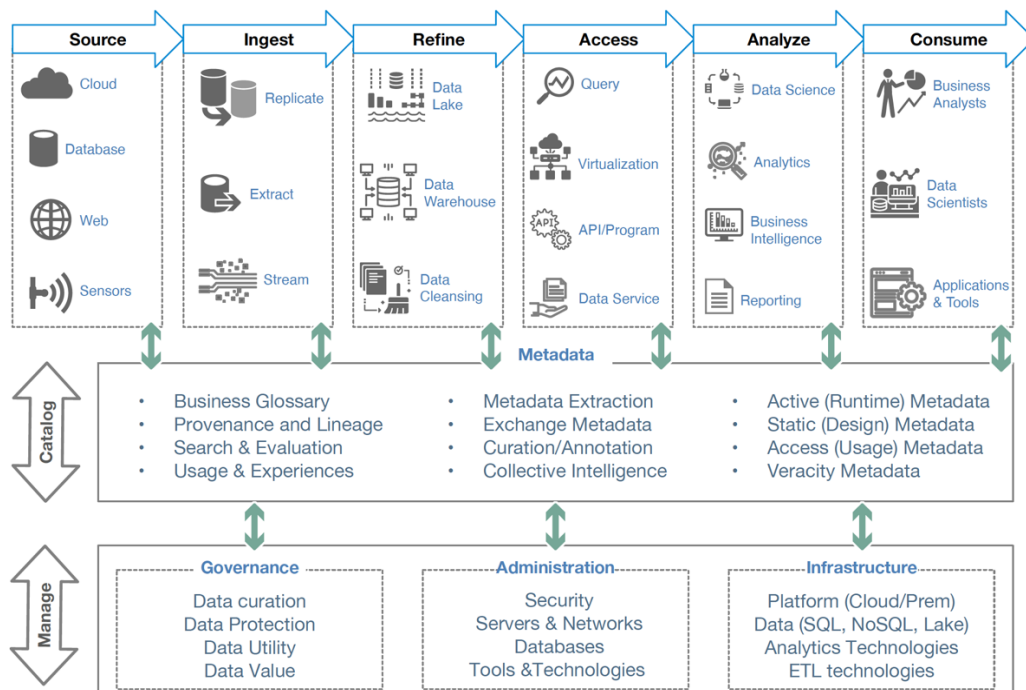


Figure 2. Contextual Data Fabric – Core Components and Features

The CDF used in this application ingests direct measurements from multiple best-in-class detection sources (see Sections 10 and 11) to deliver the most accurate digital representation of ground truth asset operations performance and their ambient environmental impact.

Immutable Data Platform

The Immutably™ Data Platform is a CDF that specializes in multi-source ingestion and digital processing. The system can manage an arbitrary number of oracles that augment the data payload by submitting additional metadata facts to describe the context, as well as notaries that consist of qualified third parties and subject matter experts (“SME”) that can attest to the veracity of the data payload through digitally signed statements. The Immutably™ Data Platform creates an infrastructure that provides the ability to trace acquired data (traceability) back to the point of origination. To ensure complete auditability, all data within Immutably is recorded to a distributed ledger (blockchain).

As previously mentioned, verification of data is performed through Digital Notaries. A Digital Notary is an entity, individual, or group of individuals with relevant expertise and purview to attest to the quality of data generated. In the context of environmental/emissions monitoring and measurement, Digital Notaries could include (not limited to):

- Emissions Performance Testing Vendors
- LDAR Technicians
- Satellite Data Analysts
- External 3rd Party Verifiers

Notaries are responsible for providing artifacts that support the validity and integrity of the acquired data/measurements. It is possible to apply additional algorithms to “weight” the efficacy of each data feed based on established criteria surrounding each data point. For example – if a Notary cannot attest to the validity of a certain set of data points, then the resulting data points could be down-weighted—or disposed of altogether if the measurements are unreliable. Digital Notarization can be performed by human beings or algorithmically through executable programs, depending on the specific application.

Distributed Ledger and Identity

At the heart of the CDF is the notion of a trusted identity. All data and entities within the system must be assigned a unique, unforgeable identity used as the basis for linkage, but also non-repudiation via digital signatures. Traditionally, Public Key Infrastructure (PKI) has relied upon unsafe propagation of certificates and identities, but in Immutably, this is done using distributed ledger technology.

Appendix B – Revision Log

Version Number	Date	Revisions	Made by
2.0	December 2022	Issued for Use	Matt Berchtold
2.1	February 2024	- CO₂ and Scope 2 calculation equations updated - For combustion sources, differentiated between survey requirements for engines, turbines, and fired heaters - Specified leak duration calculations for equipment leaks - Included comments to allow use of measured leak rates for malfunctioning pneumatics - Included comments around use of engineering estimates for flare calculations and dry seal vents - Added crankcase venting as an emissions source	Priya Germer
2.2	April 2024	Added qualitative data quality indicators & exception reporting	Priya Germer
2.3	May 2024	General clean-up on language, including eliminating references to specific sensor manufacturers	Matt Berchtold
2.4	August 2024	Added statistically representative sampling	Jared Marshall
2.5	May 2026	- Separated and added more detail to Scope 2 emissions - Added in CO₂e calculations - Added NGL support - Added Natural Gas Fired Power Generation support - Updated various table and equation references to account for external updates (e.g. EPAGHG RP) - Added maintenance workovers emission source. - Methodology title modified - Updated Verification requirements	Jared Marshall